

**GANDHI INSTITUTE OF
ENGINEERING AND TECHNOLOGY
UNIVERSITY, ODISHA, GUNUPUR**
(SCHOOL OF ENGINEERING & TECHNOLOGY)



BACHELOR'S DEGREE PROGRAMME
(B. Tech – Computer Science and
Engineering)

Course Structure and Syllabus

Academic Session

[2023-27]



GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY, ODISHA, GUNUPUR

SCHOOL OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

1. Institute Vision & Mission, Department Vision & Mission, PEO's, PO's, PSO's

Institute Vision:

To be a renowned and Globally Recognized University giving importance to academic excellence with latest technology, Research, Innovation and Entrepreneurial Attitude.

Institute Mission:

- To create innovative and committed work force to cater to the societal, environmental and economic needs of the Nation.
- To promote Education and Research Globally at par with the international standards
- To prepare the future leaders with latest skills befitting to become Entrepreneurs or Employable.
- To support and uplift the meritorious students of this Tribal Area to represent as an Ambassador in all forums as a part of our social responsibility.

Department Vision:

To be globally acclaimed as a premier Computer Science and Engineering department through excellent teaching and strong research environment by creating professionally competent engineers with moral values committed to build a vibrant nation.

Department Mission:

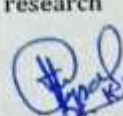
- Provide a positive and professional learning environment to master the fundamental principle of computing, problem solving skills, leadership qualities, team-spirit, ethics and societal responsibilities.
- To enable the graduates to use modern tools and technologies in the area of computer science and information technology.
- Establish Industry Institute Interaction program to enhance the employability skills.
- To promote innovative research in Computer Science and Engineering to serve the needs of Industry, Government and Society at large.

Program Educational Objectives:

- To render best platform for adequate training and opportunities to work as teams on projects with effective communication skills and leadership qualities and understand professional ethics, social awareness and organizational context in which their engineering skills are utilized.

To endow the students with sound knowledge in the field of mathematics, basic science and engineering fundamentals to solve and inculcate the ability to utilize their skills to prepare them for higher studies, research and analyze engineering problems.




R. MURALI GOPAL
MOD. CSE
GIET UNIVERSITY
GUNUPUR-765022

GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY, ODISHA, GUNUPUR

SCHOOL OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

- To extend an ability to analyze the need of the society by providing innovation solutions, leading to their personal cum professional growth as an entrepreneur.

Program Outcome:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY, ODISHA, GUNUPUR
SCHOOL OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

- 11. Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- 12. Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes:

- To provide students an understanding of the expectations of industry and practical competence with a broad range of programming language and open source platforms through value added courses.
- The ability to analyze and develop computer programs in the areas related to Artificial Intelligence, Big Data Analysis and Cyber Security for efficient design of computer-based systems of varying complexity.

Program Educational Objectives:

- Develop proficiency in the field of mathematics, basic science, and engineering fundamentals to analyze and solve a wide range of computational problems in industry or other work environments and succeed in an advanced degree program in a field such as engineering.
- Possess the ability to collaborate as team members and team leaders for facilitating technical solutions for computing systems with awareness and commitment to their ethical responsibilities.
- Attain the ability to critically analyze the need of society and provide innovative solutions in multi-disciplinary areas adopting technologies leading to their personal and professional growth as an entrepreneur.

III Semester [Second Year]
B. Tech in Computer Science and Engineering (2023 Admission)

SN	CC	Sub Code	Name of the Subject	L	T	P	C	ISA	ESA
THEORY									
1	BS	23BCSBS23001	Discrete Mathematics	3	1	0	4	40	60
2	ES	23BCSES23001	Digital Electronics Circuits	3	0	0	3	40	60
3	PC	23BCSPC23001	Database Management Systems	3	0	0	3	40	60
4	PC	23BCSPC23002	Object Oriented Programming through Java	3	0	0	3	40	60
5	PE	23BCSPE23001	PE-I (From Bucket)	3	0	0	3	40	60
6	HS	23BBSHS23001	Intermediate communication skills and critical thinking	1	0	0	1		
			TOTAL	16	1	0	17	200	300
PRACTICAL/SESSIONAL									
7	ES	23BCSES23101	Digital Electronics Circuits Lab	0	0	2	1	30	20
8	PC	23BCSPC23102	Object Oriented Programming through Java Lab	0	0	2	1	30	20
9	PC	23BCSPC23101	Data Base Management Systems Lab	0	0	2	1	30	20
10	HS	23BBSHS23101	Intermediate communication skills and critical thinking lab	0	0	2	1	30	20
11	PST	23BCSPW23101	Summer Internship-I	0	0	0	1	30	20
12	MC	23BCSLP23101	Learning Project-I	0	0	0	1	50	0
			TOTAL	0	0	8	6	200	100
			TOTAL	16	1	8	23	400	400

Professional Elective Bucket

Sl.	Sem	Professional Elective I
1	23BCDPC23011	Introduction to Data Science
2	23BCDPC23012	Data Preparation and Analysis

Subject Code	Name of the Subject	L	T	P	C	
	Discrete Mathematics	3	1	0	4	
Course Educational Objectives						
CEO1	To know logical equivalence and to apply in various proofs.					
CEO2	To know about relation and recurrence relation.					
CEO3	To study Boolean Algebra and its properties.					
CEO4	To know about Graphs and trees.					
Course Outcomes: Upon successful completion of this course, students should be able to:						
CO1	Understanding the concept of logic, groups, lattice. graphs and trees					
CO2	Memorize the logical gates, recurrence formulas and different types of graphs					
CO3	Evaluate the arguments, predicates and quantifiers, solution of recurrence relations, methods of solution					
CO4	Analyze recurrence relation by generating functions and groups, binary trees					
CO5	Categorize different types of graphs, and its property					
CO6	Solve the logical equivalences, Boolean algebra and pre order and post order of arithmetic Operations.					

CO-PO & PSO Mapping

COs	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	1	2													
CO2	2	3													
CO3	2	3													
CO4	2	3													
CO5	2	3													
CO6	2	3													
Avg.	1.8	2.8													

SYLLABUS

UNIT I - MATHEMATICAL LOGIC AND SET THEORY (12 Hours)

Propositional logic, Propositional Equivalence, Predicate Calculus, and Quantifiers Statement Calculus, Nested Quantifiers, Proof methods and Strategies, Mathematical Induction

UNIT II- RECURRENCE RELATION (12Hours)

Recurrence relation, Solution to recurrence relation, Generating functions, Inclusion and exclusion principle Relation and their properties, Closure of relations, Equivalence relations, Partial order relations, posets

UNIT III - BOOLEAN ALGEBRA AND LATTICES (12 Hours)

Lattices, Distributive and Complemented Lattices, Sub-lattices, Boolean Lattices and Boolean algebra, Boolean Functions and Boolean Expressions

UNIT IV- GROUP THEORY (12 Hours)

Semi groups, Monoids, Groups, Subgroups, and Permutation groups, Normal subgroups, Homomorphism, isomorphism, Kernel.

UNIT V- GRAPH THEORY (12 Hours)

Basic Definitions – Some Special Graphs – Matrix Representation of Graphs -- Paths and circuits – Eulerian Theorem, Hamiltonian Theorem, Graphs – connected graphs, Graph coloring, Trees - Spanning Trees - Rooted trees – Binary Trees, Minimum Spanning tree -Kruskal's algorithm, Prim's algorithm, Tree Traversal.

Reference Books:

1. Kenneth H. Rosen, “Discrete Mathematics and its Applications”, Sixth Edition, 2008, Tata McGraw Hill Education, New Delhi
2. C. L. Liu and D. Mohaptra, “Elements of Discrete Mathematics”, Third Edition, 2008,
3. N. Deo, Graph Theory and Applications to Engineering and Computer Science, Prentice Hall of India
4. Discrete Mathematics by Schaum’s Outlines(Second Edition)
5. Ralph P. Grimaldi,” Discrete and Combinatorial Mathematics”, Fifth Edition, 2005, Pearson Education, New Delhi.

Subject Code		Name of the Subjects										L	T	P	C
		Object Oriented Programming Through Java										3	0	2	4
Course Educational Objectives															
CEO 01	The model of object-oriented programming: abstract data types, encapsulation, inheritance and polymorphism														
CEO 02	Fundamental features of an object-oriented language like Java: object classes and interfaces, exceptions and libraries of object collections														
CEO 03	How to take the statement of a business problem and from this determine suitable logic for solving the problem; then be able to proceed to code that logic as a program written in Java.														
CEO 04															
Course Outcomes: Upon successful completion of this course, students should be able to:															
CO 01	Knowledge of the structure and model of the Java programming language.														
CO 02	Use the Java programming language for various programming technologies.														
CO 03	Develop software in the Java programming language.														
CO 04	Evaluate user requirements for software functionality required to decide whether the Java programming language can meet user requirements.														
CO 05	Propose the use of certain technologies by implementing them in the Java programming language to solve the given problem.														
CO 06	Choose an engineering approach to solving problems, starting from the acquired knowledge of programming.														
CO-PO & PSO Mapping (Articulation Matrix):															
COs	PROGRAMME OUTCOMES												PSOs		
	01	02	03	04	05	06	07	08	09	10	11	12	01	02	
01	3	1											1		
02	3	2											1		
03	2	2	2										2		
04	2	1	2										1		
05	2	1	3										1		
06	1	3	3										1		
Avg.	2.1	1.6	2.5										1.16		
Syllabus															
UNIT I												13 Hours			
An introduction to Object Oriented Programming, Features of Object Oriented Programming Introduction to Java. Difference between C/C++ and Java, Features of Java, First Java Program, Writing the java program, Compiling the program, JVM and its significance in executing a program, Architecture of JVM. Taking Input from keyboard, Command Line Arguments, Using Scanner Class, Using Buffered Reader class. Java Tokens, Datatypes, Operators, Control Structures, Conditional Statements, Iteration Statement/Loops, Jumping Statements, Java Arrays, 1-dimensional arrays, 2-dimensionanl arrays.															
UNIT II												15 Hours			
Introduction to Classes and Objects. Constructors, static Keyword, this Keyword, Array of Objects, Access Modifiers (Public, Private, Protected, Default). Inheritance, Types of Inheritance and Java-supported Inheritance, super, Polymorphism, Method Overloading, Constructor Overloading, Method Overriding, Dynamic Method Dispatching. Interfaces, Multiple Inheritance Using Interfaces, abstract classes. String Handling in Java: String, StringBuffer, StringBuilder, StringTokenizer Wrapper classes, Auto boxing and unboxing.															

UNIT III		12 Hours
Java API Packages, User-Defined Packages, Accessing Packages. Error and Exception Handling, Types of exceptions Hierarchy of Exception classes, try, catch, finally, throw, throws, Commonly used Exceptions and their details ,User defined exception classes. Multithreading, Thread in Java, Thread execution prevention methods. (yield(), join(), sleep()), Concept of Synchronization, Inter Thread Communication, Basics of Deadlock, Demon Thread. Inner Classes: Introduction, Member inner class, Static inner class, Local inner class, Anonymous inner class. IO Streams (java.io package), Byte Stream and Character Stream, Files and Random Access Files, Serialization.		
UNIT IV		12 Hours
Collection Frame Work (java.util), Util Package interfaces, List, Set, Map. Applet Introduction, Life Cycle of an Applet, GUI with an Applet. AWT : The AWT class hierarchy, user interface components- Labels, Button, Text Components, Check Box, Check Box Group, Choice, List Box, Panels – Scroll Pane, Menu, Scroll Bar. Working with Frame class, Colour, Fonts and layout managers. Event Handling: Events, Event sources, Event Listeners, Event Delegation Model (EDM), Handling Mouse and Keyboard Events, Adapter classes		

Subject Code		Name of the Subject						L	T	P	C				
		Database Management System						3	0	1	4				
Course Educational Objectives															
Pre -Requisite: A student should have basic idea on programming and file systems.															
CEO 1	Gain a good understanding of the architecture and functioning of Database Management Systems as well as associated tools and techniques.														
CEO 2	Understand and apply the principles of data modeling using Entity Relationship and develop a good database design.														
CEO3	Understand the use of Structured Query Language (SQL) and its syntax														
CEO4	Apply Normalization techniques for effective database design.														
Course Outcomes: <i>Upon successful completion of this course, students should be able to:</i>															
CO1	Identify and Classify the concepts of Database Management system, Data models and architecture of database, ER to Relational mapping concepts.														
CO2	Applying The constraints in database using different query languages like:- relational algebra and calculus implementing the Data definition and data manipulate languages in Database.														
CO3	Compare the different normal forms to Apply normalization process to construct consistent Database.														
CO4	Design and Develop the Database with concurrency control and recovery strategies to make complete Database without confliction and anomalies in concurrent access environment.														
CO5	Develop efficient storage scheme of saving and retrieving Records and Files														
CO6	Apply the principles of database transaction management, database recovery and security.														
CO-PO & PSO Mapping:															
COs	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	3	1	1												
CO2	3	2	1												
CO3	2	1	1												
CO4	2	1	2												
CO5	2	1	2												
CO6	3	2	1												
Avg.	2.5	1.33	1.33												
SYLLABUS															
UNIT:1 (6 Hours)															
Introduction to database Systems: advantages of database system over traditional file system, Basic concepts & Definitions, Database users, Database Language, Database System Architecture, Schemas, Sub Schemas, & Instances, database constraints, 3-level database architecture, Data Abstraction, Data Independence, Mappings, Structure, Components & functions of DBMS, Data models.															
Entity-Relationship model: Basic concepts, Design process, Constraints, Keys, Design issues, E-R Diagrams, weak entity sets, Extended E-R features – generalization, specialization, aggregation, Reduction to E-R database schema.															

UNIT:2	(10 Hours)
Relational Query Languages: Relational Operations. Relational Algebra, Selection and projection set operations, renaming, Joins, Division, Examples of Algebra overviews, Relational calculus, Tuple relational Calculus, Domain relational calculus, Extended relation algebra operation	
UNIT:3	(10 Hours)
Normalization: Introduction, non-loss decomposition and functional dependencies, First, Second, and third normal forms – dependency preservation, Boyce-Codd normal form. Higher Normal Forms - Introduction, Multi-valued dependencies and fourth normal form, Join dependencies and fifth normal form	
UNIT:4	(10 Hours)
Transaction Concept: Transaction properties (ACID), Transaction State, Schedules, Serializability (Conflict Serializability and View Serializability), Need for Concurrency, Concurrency Control- Lock based control, time stamping Protocol, Two Phase Locking, Dead lock in DBMS	
Recovery and Atomicity: Log Based Recovery, Transaction Logs and Shadow Paging, Types of Recovery Techniques in DBMS, Rollback/Undo Recovery Technique, Commit/Redo Recovery Technique, Checkpoint Recovery, Backup Techniques	
UNIT:5	(6 Hours)
Detailed Storage Architecture: Storage Strategies: Detailed Storage Architecture, Storing Data, Magnetic Disk, RAID, Other Disks, Magnetic Tape, Storage Access, File & Record Organization, File Organizations & Indexes, Order Indices, B+ Tree Index Files, Hashing Data Dictionary	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books:	
1. Sudarshan, Korth: Database System Concepts, 6th edition, McGraw-Hill Education 2. Elmasari & Navathe: Fundamentals of Database System, Pearson Education.	
Reference Books:	
1. Elmasari & Navathe: Fundamentals of Database System, Pearson Education. 2. Ramakrishnan: Database Management Systems, McGraw-Hill Education. 3. Andrew S. Tanenbaum: Modern Operating Systems, 3rd Edition, Pearson Education. 4. Terry Dawson, Olaf Kirch: Linux Network Administrator's Guide, 3rd Edition O'Reilly	

Subject Code		SUBJECT NAME											L	T	P	C
		Introduction to Data Science											3	0	0	3
Course Educational Objectives																
CEO 01	Able to apply fundamental algorithmic ideas to process data															
CEO 02	Able to construct predictive models to classify new data set															
CEO 03	Learn to apply hypotheses and data into actionable predictions															
CEO 04	Document and communicate the results effectively to different stakeholders and effectively communicate the findings using visualization techniques.															
Course Outcomes: : Upon successful completion of this course, students should be able to:																
CO1	Understand the Foundations of Data Science. Conduct Data Acquisition and Pre-processing															
CO2	Describe the data using various statistical measures															
CO3	Work Collaboratively in Data Projects. Use Relevant Data Science Tools and Technologies															
CO4	Apply Data Modelling Techniques. Evaluate and Select the appropriate Model.															
CO5	Perform data reduction and apply visualization techniques															
CO6	Apply Data Science Skills to Real-World Problems. Continuously Learn and Stay Updated															
CO-PO & PSO Mapping (Articulation Matrix):																
COs		PROGRAMME OUTCOMES												PSOs		
	01	02	03	04	05	06	07	08	09	10	11	12	01	02		
01	3	2	-	-	-	-	-	-	-	-	-	-	-	-		
02	2	3	3	-	2	-	-	-	-	-	-	-	-	-		
03	2	3	2	-	2	-	-	-	2	-	-	-	-	-		
04	2	1	-	-	-	2	1	-	-	-	-	2	-	-		
05	2	2	3	-	-	-	-	-	-	-	-	-	-	-		
06	1	-	-	-	-	-	-	-	-	-	-	2	-	-		
Avg.	2	2.2	2.6		2	2	1		2			2				
Syllabus																
UNIT I												(12 Hours)				
Introduction to Data Science – Evolution of Data Science – Data Science Roles – Stages in a Data Science Project – Applications of Data Science in various fields – Data Security Issues, Data Collection Strategies, Data Categorization: NOIR Topology.																
UNIT II												(10 Hours)				
Descriptive Statistics – Mean, Median, Standard Deviation, Skewness and Kurtosis. Data Pre-Processing Overview – Data Cleaning – Data Integration and Transformation – Data Reduction – Data Discretization.																
UNIT III												(8 Hours)				
Model Selection and Evaluation using Visualization: - Simple and Multiple Regression, Polynomial Regression, Ridge Regression, Box Plots, Residual Plot, Distribution Plot, Pivot Table.																
UNIT IV												(10 Hours)				
Introduction hypothetical testing: -NULL(H ₀) hypothesis and Alternative hypothesis, Type-I and Type-II error, Test statistics:- Z-statistics, T-statistics, Chi-Square Test, Correlation Statistics – ANOVA, correlation Analysis- Heat Map.																
UNIT V												(6 Hours)				
Model evaluation Metrics: - Regression metrics and classification metrics– Cross Validation – Overfitting – Under Fitting and Testing Multiple Parameters by using Grid Search.																
Teaching Methods: Chalk& Board/ PPT/Video Lectures/Lecture by Industry Expert/MOOCs																

Text Books:

1. Cathy O'Neil and Rachel Schutt , *"Doing Data Science"*, O'Reilly, 2015.
2. Cathy O'Neil and Rachel Schutt , *"Doing Data Science"*, O'Reilly, 2015.

Reference Book:

1. Jojo Moolayil, *"Smarter Decisions : The Intersection of IoT and Data Science"*, PACKT, 2016.
2. Raj, Pethuru, *"Handbook of Research on Cloud Infrastructures for Big Data Analytics"*, IGI Global

		DIGITAL ELECTRONICS	3-0-0
Pre-requisites (if any):		Knowledge of Basic Mathematics & Basic Electrical and Electronics Engineering.	
Course Objectives:			
	To understand digital logic fundamentals, including binary systems and Boolean algebra.		
	To design and analyze combinational and sequential circuits, including logic gates, flip- flops, and counters.		
	To apply digital electronics in real-world applications, using simulation tools and programmable logic devices (PLDs).		
Course Content			
Unit 1		(10 Hours)	
Number Systems and Codes: Binary, Octal, Hexadecimal, and Decimal Number System and their Conversion; Representation of Signed Binary and Floating-Point Number; Binary Arithmetic using 1’s and 2’s Complements, Binary Codes - BCD Code, Gray Code, ASCII Character Code.			
Boolean Algebra and Logic Gates: Axioms and Laws of Boolean Algebra; Reducing Boolean Expressions; Logic Levels and Pulse Waveforms; Logic Gates; Boolean Expressions and Logic Diagrams; Canonical and Standard Forms.			
Unit 2		(12 Hours)	
Gate-level Minimization: K-maps - Two, Three, and Four Variable K-maps, Don’t-Care Conditions; NAND and NOR Implementation; Other Two-Level Implementations, Exclusive-OR Function.			
Combinational Logic: Combinational Circuits; Analysis Procedure; Design Procedure; Adders; Subtractors; Parallel Binary Adders; Binary Adder-Subtractor; Binary Multiplier; Magnitude Comparator; Decoders; Encoders; Multiplexers; De-multiplexers.			
Unit 3		(8 Hours)	
Synchronous Sequential Logic: Sequential Circuits; Storage Elements: Latches, Flip-Flops, Master-Slave Flip-Flop; Conversion of Flip-Flops; Analysis of Clocked Sequential Circuits; Mealy and Moore Models of Finite State Machines; Design Procedure.			
Unit 4		(8 Hours)	
Registers and Counters: Shift Registers; Data Transmission in Shift Registers; SISO, SIPO, PISO, and PIPO Shift Registers; Counters; Asynchronous Counters; Design of Asynchronous Counters; Synchronous Counters; Design of Synchronous Counters; Ring Counter.			
Unit 5		(6 Hours)	
Memory and Programmable Logic: Introduction; Random-Access Memory; Memory Decoding; Error Detection and Correction; Read-Only Memory; Programmable Logic Array; Programmable Array Logic; Sequential Programmable Devices.			
Text Books:			
[1] M. Morris Mano, and Michael D. Ciletti, <i>Digital Design: With an Introduction to the Verilog HDL, VHDL, and SystemVerilog</i> . Pearson Education, Sixth Edition, 2017.			

- | | |
|-------------------------------------|--|
| <p>[2]</p> <p>[3]</p> | <p>A. Anand Kumar, <i>Fundamentals of Digital Circuits</i>. PHI Learning Pvt. Ltd., New Delhi, Fourth Edition, 2016.</p> <p>R. P. Jain, <i>Modern Digital Electronics</i>. Tata McGraw-Hill Education Pvt. Ltd., Fourth Edition, 2009.</p> |
|-------------------------------------|--|

Reference Books:

- [1] John P. Uyemura, *A First Course in Digital Systems Design: An Integrated Approach*. Thomson Press (India) Ltd., India Edition, 2002.
- [2] William H. Gothmann, *Digital Electronics: An Introduction to Theory and Practice*, PHI Learning Pvt. Ltd., New Delhi, Second Edition, 2006.
- [3] D.V. Hall, *Digital Circuits and Systems*. Tata McGraw-Hill Education Pvt. Ltd., 1989.
- [4] Charles H. Roth, *Digital System Design using VHDL*. Tata McGraw-Hill Education Pvt. Ltd., Second Edition, 2012.

Course Outcomes:

At the end of this course, students will be able to

CO1	Define and analyze the number systems and conversions.
CO2	Perform binary arithmetic with 1's and 2's complements and understand binary codes.
CO3	Apply Boolean algebra laws to simplify expressions and design logic circuits
CO4	Use K-maps for logic minimization and design combinational circuits.
CO5	Analyze and design synchronous sequential circuits.
CO6	Design shift registers, and counters, and understand RAM, ROM, and PLDs.

CO-PO & PSO matrices of Course

[illegible]

DIGITAL ELECTRONICS LABORATORY		0-0-2
Pre-requisites (if any):	Knowledge of Basic Electrical and Electronics Engineering.	
Course Objectives:		
–	To gain hands-on experience with basic logic gates and combinational circuits.	
–	To implement and test combinational and sequential circuits.	
–	To utilize simulation tools for circuit design and testing.	
Course Content		
List of Experiments:		
(At least 10 experiments should be done, Experiments No. 1 and 2 are compulsory, and out of the balance 8 experiments at least 3 experiments have to be implemented through both Verilog/VHDL and hardware implementation as per the choice of the student totaling to 6 and the rest 2 can be either through Verilog/VHDL or hardware implementation.)		
1. Digital Logic Gates: Investigate the logic behavior of AND, OR, NAND, NOR, EX-OR, EXNOR, Inverter, and Buffer Gates, use of Universal NAND/NOR Gate. 2. Gate-Level Minimization: Two-level and multi-level implementation of Boolean functions. 3. Combinational Circuits: Design, assemble, and test: adders and subtractors, code converters, gray code to binary, and 7-segment displays. 4. Design, implement, and test a given design example with (i) NAND Gates only, (ii) NOR Gates only, and (iii) using the minimum number of Gates. 5. Design with Multiplexers and De-multiplexers. 6. Flip-Flop: Assemble, test, and investigate the operation of SR, D & JK flip-flops. 7. Shift Registers: Design and investigate the operation of all types of shift registers with parallel load. 8. Counters: Design, assemble, and test various ripple and synchronous counters - decimal counter, Binary counter with the parallel load. 9. Memory Unit: Investigate the behavior of the RAM unit and its storage capacity - 16×4 RAM: testing, simulating, and memory expansion. 10. Clock-Pulse Generator: Design, implement, and test. 11. Parallel Adder and Accumulator: Design, implement, and test. 12. Binary Multiplier: Design and implement a circuit that multiplies 4-bit unsigned numbers to produce an 8-bit product. 13. Verilog/VHDL simulation and implementation of Experiments listed in Sl. No. 3 to 12.		

Subject Code		Name of the Subject	L	T	P	C	
		Intermediate Communication Skills and Critical Thinking	3	0	0	3	
Course Educational Objectives							
CEO1	To develop the communication skills and soft skills of the students.						
CEO2	To enhance employability and entrepreneurial skills.						
CEO3	To promote critical thinking.						
Course Outcomes: Upon successful completion of this course, students should be able to:							
CO1	Recognise the importance of LSRW skills.						
CO2	Develop effective listening skills.						
CO3	Demonstrate public speaking skills.						
CO4	Distinguish professional writing from general writing.						
CO5	Develop effective reading skills.						
CO6	Execute critical thinking.						

CO-PO & PSO Mapping

COs	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	2														
CO2		2													
CO3	2	2													
CO4	2														
CO5	2														
CO6															
Avg.	2	2													

SYLLABUS

UNIT-1 Comprehensive Listening Skills	(4 Hours)
1.1 What and why	
1.2 Techniques for improving listening skills	
UNIT-2 Public Speaking Skills	(4 Hours)
2.1 Public speaking; what and why	
2.2 Overcoming stage fear	
2.3 Techniques for effective speaking skill	
UNIT- 3 Stages of Writing	(4 Hours)
3.1 Pre-Writing- selecting a topic	
3.2 Writing- Free Writing, Drafting	
3.3 Re-writing- Revision, editing, proof Reading	
UNIT- 4 Effective Reading Skills	(3 Hours)
4.1 What is Reading Comprehension	
4.2 Tips for effective reading	
4.3 Techniques and types.	
UNIT-5 Critical Thinking Skills	(4 Hours)
5.1 What and Why	
5.2 Critical thinking vs. Lateral thinking	
5.3 Ways to improve Critical Thinking	

Teaching Methods: Chalk & Board/PPT/Video Lectures

Reference Books:

- | |
|--|
| <ol style="list-style-type: none">1. Basic Communication Skills for Technology by Andrea J. Rutherford, Pearson Education Asia, Patparganj, New Delhi, 2001.2. How to Read better and Faster by Norman Lewis, 4th Edition, Publisher: Crowell, 1978.3. The Kabuliwala and other stories by Rabindranath Tagore, Maple Classics, 2019.4. The Gift of Magi by O Henry, Candlewick Press, 2008. |
|--|

Activity- Answering hypothetical question in a mock interview	
Video Resume	(2 hours)
Activity- Preparing a digital resume	
Professional e-mail	(2 hours)
Activity- Writing e-mail for professional purpose	
Minutes of the Meeting	(2 hours)
Activity- Preparing minutes of mock meetings.	
Memo Writing	(2 hours)
Writing Memos for internal organization	
Vocabulary Exercise	(2 hours)
Activity- Playing word games in small groups.	
Intermediate Reading Comprehension	(2 hours)
Activity-Reading an Intermediate level passage/ poems and answer questions.	
Preparing a Sales Pitch	(2 hours)
Activity- Prepare a presentation on any product (real/imaginary) of choice and present a sales pitch	
Critical Thinking Activity	(2 hours)
Activit- critically analysis of Kabuliwala by Rabindranath Tagore OR The Gift of Magi by O Henry	
Critical Thinking	(2 hours)
Activity 2- Critically Analyse Stopping by the Woods in a Snowy Evening by Robert Frost OR Ozymondias by P.B Shelly.	
Reference Books 1. An Introduction to Professional English and Soft Skills by B. K. Das et al., Cambridge University Press, 2009. 2. Communicative English for Engineers and Professionals by Nitin Bhatnagar and Mamta Bhatnagar. Published by DK/Pearson, 2010. 3. Practical English Usage . Michael Swan, OUP, 1995	

IV Semester [Second Year]
B. Tech in Computer Science and Engineering (2023 Admission)

SN	CC	Sub Code	Name of the Subject	L	T	P	C	ISA	ESA
THEORY									
1	PC	23BCSPC24001	Fundamentals of Python Programming	3	0	0	3	40	60
2	PC	23BCSPC24002	Computer Organization & Architecture	3	0	0	3	40	60
3	PC	23BCSPC24003	Design and Analysis of Algorithms	3	0	0	3	40	60
4	PC	23BCSPC24004	Operating System	3	0	0	3	40	60
5	PE	23BCSPE24001	PE-II (From Bucket)	3	0	0	3	40	60
6	HS	23BBSHS24001	English Learning -II (Advance communication skills and professional Ethics)	1	0	0	1		
			TOTAL	16	0	0	16	200	300
PRACTICAL/SESSIONAL									
7	PC	23BCSPC24101	Fundamentals of Python Programming Lab	0	0	2	1	30	20
8	PC	23BCSPC24103	Design and Analysis of Algorithms Lab	0	0	2	1	30	20
9	PC	23BCSPC24105	Unix and Shell Programming Laboratory	0	0	2	1	30	20
10	HS	23BBSHS24101	English Learning -II (Advance communication skills and professional Ethics lab)	0	0	0	1	30	20
11	PST	23BCSLP24101	Learning Project-II	0	0	0	1	50	0
			TOTAL	0	0	6	5	170	80
			TOTAL	16	0	6	21	370	380

Professional Elective Bucket

Sl.	Sem	Professional Elective II
1	23BCDPE24011	Introduction to Soft Computing
2	23BCDPE24012	Fuzzy Logic & Applications

Subject Code		Name of the Subject								L	T	P	C		
		Computer Organization and Architecture								3	0	0	3		
Course Educational Objectives															
CEO1	Identify the functional units in a digital computer system,														
CEO2	Distinguish between the various ISA styles and trace the execution sequence of an instruction through the processor.														
CEO3	Compare different approaches used for implementing a functional unit and evaluate different computer systems based on performance metrics.														
CEO4	Provide an outline of working principles of components of computer.														
Course Outcomes: : <i>Upon successful completion of this course, students should be able to:</i>															
CO1	Explain and illustrate the execution cycle of a program.														
CO2	Classify the components of digital computer and designing of control and memory units with performance evaluation.														
CO3	Compute different types of binary arithmetic operation. .														
CO4	Identify the characteristics of memories and explain memory and IO devices working process.														
CO5	Classify the concept of Parallelism														
CO6	Explain memory and types and interrupts														
CO-PO & PSO Mapping															
COs	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	1	1			2										
CO2	2	3			3										
CO3	3	2			2										
CO4	1	1			2										
CO5	2	2			1										
CO6	2	3			2										
Avg.	2	2			2										
SYLLABUS															
UNIT:1 (12 Hours)															
Functional Units of a computer, Operational concepts, Bus structures, Memory locations and addresses, Memory operations, Instructions, Instruction sequencing, addressing modes, Problem solving, Introduction to Microprocessor, Introduction to Assembly language, writing of assembly language programming, ARM Processor: The thumb instruction set, Processor and CPU cores, Instruction Encoding format, Memory load and Store instruction in ARM, Basics of IO operations.															
UNIT: 2 (12 Hours)															
Addition and subtraction of Signed numbers, Problem solving, Design of fast adders, Ripple carry adder and Carry look ahead adder, Multiplication of positive numbers, Problem Solving, Signed operand multiplication, Problem solving, Fast multiplication- Bit pair recoding of Multipliers, Problem Solving, Carry Save Addition of summands - Problem Solving, Integer division – Restoring Division - Solving Problems, Non-Restoring Division- Solving Problems, Floating point numbers and operations- Solving Problems.															

UNIT:3	(12 Hours)
Fundamental concepts of basic processing unit, Performing ALU operation, Execution of complete instruction, Branch instruction, Multiple bus organization, Hardwired control, Generation of control signals, Micro-programmed control, Microinstruction, Micro-program Sequencing, Micro instruction with Next address field, Basic concepts of pipelining, Pipeline Performance, Pipeline Hazards-Data hazards, Methods to overcome Data hazards, Instruction Hazards, Hazards on conditional and Unconditional Branching, Control hazards, Influence of hazards on instruction sets.	
UNIT:4	(12 Hours)
Parallelism, Need, types of Parallelism, applications of Parallelism, Parallelism in Software, Instruction level parallelism, Data level parallelism, Challenges in parallel processing, Architectures of Parallel Systems - Flynn's classification, SISD, SIMD, MIMD, MISD, Hardware multithreading, Coarse Grain parallelism, Fine Grain parallelism, Uni-processor and Multiprocessors, Multi-core processors, Memory in Multiprocessor Systems, Cache Coherency in Multiprocessor Systems, MESI protocol for Multiprocessor Systems.	
UNIT:5	(10 Hrs)
Memory systems -Basic Concepts, Memory hierarchy, Memory technologies, RAM, Semiconductor RAM, ROM, Types, Speed, size cost, Cache memory, Mapping Functions, Replacement Algorithms - Problem Solving, Virtual Memory, Performance considerations of various memories, Input Output Organization, Need for Input output devices, Memory mapped IO, Program controlled IO, Interrupts-Hardware, Enabling and Disabling Interrupts, Handling multiple Devices.	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books: 1. Carl Hamacher, Zvonko Vranesic, Safwat Zaky and Naraig Manjikian, "Computer Organization and Embedded Systems", Sixth Edition, Tata McGraw Hill, 2012. 2. David A. Patterson and John L. Hennessy, "Computer Organization and Design: The Hardware/Software Interface", Fourth Edition, Morgan Kaufmann / Elsevier, 2009. 3. Kai Hwang and F.A. Briggs, "Computer Architecture and Parallel Processing", McGraw Hill.	
Reference Books: 4. M. Morris Mano, "Computer System Architecture", PHI 5. William Stallings, "Computer Organization and Architecture – Designing for Performance", Sixth Edition, Pearson Education, 2003. 6. John P. Hayes, "Computer Architecture and Organization", Third Edition, Tata McGraw Hill, 1998. 7. John L. Hennessy and David A. Patterson, "Computer Architecture – A Quantitative Approach", Morgan Kaufmann / Elsevier Publishers, Fifth Edition, 2012.	

Subject Code	Name of the Subject	L	T	P	C										
	Design and Analysis of Algorithms	3	0	0	3										
Course Educational Objectives															
CEO1	Analyze the asymptotic performance of algorithms														
CEO2	Demonstrate familiarity with major algorithms														
CEO3	Apply important algorithmic design paradigms and methods of analysis														
CEO4	Synthesize efficient algorithms in common engineering design situations														
Course Outcomes: Upon successful completion of this course, students should be able to:															
CO1	Apply efficient algorithms to reduce space and time complexity of both recurrent and														
CO2	Solve problems using divide and conquer approaches														
CO3	Apply greedy and dynamic programming types techniques to solve polynomial time														
CO4	Create exponential problems using backtracking and branch and bound approaches														
CO5	Interpret various approximation algorithms and interpret solutions to evaluate P														
CO6	Create algorithms that are efficient in space and time complexities by using divide														
CO-PO & PSO Mapping															
COs	PROGRAMMEOUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	1	1	1												
CO2	2	2	2												
CO3	3	1	1												
CO4	3	1	1												
CO5	2	2	3												
CO6	1	3	2												
Avg.	2	1.6	1.6												
SYLLABUS															
Unit I (15Hrs)															
Introduction-Algorithm Design, Fundamentals of Algorithms, Correctness of algorithm, Time complexity analysis, Insertion sort-Line count, Operation count, Algorithm Design paradigms, Designing an algorithm, And its analysis-Best, Worst and Average case, Asymptotic notations Based on growth functions, O,O,Θ, ω, Ω, Mathematical analysis, Induction, Recurrence relations- Solution of recurrence relations, Substitution method- Solution of recurrence relations, Recursion tree- Solution of recurrence relations.															
Unit -II (15Hrs)															
Introduction-Divide and Conquer, Maximum Subarray Problem, Binary Search, Complexity of binary search, Merge sort, Time complexity analysis, Quick sort and its Time complexity analysis, Best case, Worst case, Average case analysis, Strassen's Matrix multiplication and its recurrence relation, Time complexity analysis of Merge sort, Largest sub-array sum, Time complexity analysis of Largest subarray sum, Master Theorem Proof, Master theorem examples, Finding Maximum and Minimum in an array, Time complexity analysis-Examples, Algorithm for finding closest pair problem, Convex Hull problem.															
Unit – III (10Hrs)															
Introduction-Greedy and Dynamic Programming, Examples of problems that can be solved by using greedy and dynamic approach, Huffman coding using greedy approach, Comparison of brute force and Huffman method of encoding, Knapsack problem using greedy approach, Complexity derivation of knapsack using greedy, Tree traversals, Minimum spanning tree –															

greedy Kruskal's algorithm – greedy, Minimum spanning tree - Prims algorithm, Introduction to dynamic programming, 0/1 knapsack problem, Complexity calculation of knapsack problem, Matrix chain multiplication using dynamic programming, Complexity of matrix chain multiplication, Longest common subsequence using dynamic programming, Explanation of LCS with an example, Optimal binary search tree (OBST) using dynamic programming, Explanation of OBST with an example.	
Unit – IV	(10 Hrs)
Introduction to backtracking - branch and bound, N queen's problem – backtracking, Sum of subsets using backtracking, Complexity calculation of sum of subsets, Graph introduction, Hamiltonian circuit – backtracking, Branch and bound - Knapsack problem, Example and complexity calculation, Differentiate with dynamic and greedy, Travelling salesman problem using branch and bound, Travelling salesman problem using branch and bound example, Time complexity calculation with an example, Graph algorithms, Depth first search and Breadth first search, Shortest path introduction, Floyd-Warshall Introduction, Floyd-Warshall with sample graph, Floyd-Warshall complexity.	
Unit – V	(10 Hrs)
Introduction to randomization and approximation algorithm, Randomized hiring problem, Randomized quick sort, Complexity analysis, String matching algorithm – Examples, Rabin Karp algorithm for string matching - Example discussion, Approximation algorithm, Vertex covering, Introduction Complexity classes, P type problems, Introduction to NP type problems, Hamiltonian cycle problem, NP complete problem introduction, Satisfiability problem, NP hard problems – Examples.	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Book: 1. <i>Introduction to Algorithms</i> , T.H. Cormen, C.E. Leiserson, R.L. Rivest, C. Stein, 2nd Edition, PHI Learning Pvt. Ltd. 2. <i>Fundamentals of Algorithm</i> , Horowitz & Sahani, 2nd Edition, Universities Press.	
Reference Book: 1. <i>Algorithms, Design and Analysis</i> , H. Bhasin, First Edition, Oxford University 2. <i>Design and Analysis of Algorithm</i> , S. Sridhar, Oxford University Press 3. <i>Algorithms</i> , Sanjay Dasgupta, Umesh Vazirani, McGraw-Hill Education.	

Subject Code		Name of the Subject								L	T	P	C			
		Operating System								3	0	2	4			
Course Educational Objectives																
CE01	Provide an insight into the basic organization of computer systems.															
CE02	Familiarize students with various components of an Operating System															
CE03	Focus on fundamental problems and optimal solutions for resource management															
CE04	Understand different possible solutions for handling deadlock situation.															
Course Outcomes: <i>Upon successful completion of this course, students should be able to:</i>																
CO1	<i>Identify the need of an Operating system</i>															
CO2	<i>Know the Process management functions of an Operating system</i>															
CO3	<i>Understand the need of Memory Management functions of an Operating system</i>															
CO4	<i>Find the significance of Device management role of an Operating system</i>															
CO5	<i>Recognize the essentials of File Management part of an Operating system</i>															
CO6	<i>Gain an insight of Importance of an Operating system through practical</i>															
CO-PO & PSO Mapping																
COs	PROGRAMME OUTCOMES												PSOs			
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	
CO1	2	2	1				1									
CO2	2	1	2				2									
CO3	1	1	3				3									
CO4	2	3	2				1									
CO5	3	1	1				1									
CO6	1	2	1				1									
Avg.	1.8	1.6	1.6				1.5									
SYLLABUS																
UNIT:1 (14 Hours)																
Operating System Objectives and functions, Gaining the role of Operating systems, The evolution of operating system, Major achievements, Understanding the evolution of Operating systems from early batch processing systems to modern complex systems, OS Design considerations for Multiprocessor and Multicore, Understanding the key design issues of Multiprocessor Operating systems and Multicore Operating systems, PROCESS CONCEPT– Processes, PCB, Understanding the Process concept and Maintenance of PCB by OS, Threads – Overview and its Benefits, Understanding the importance of threads, Process Scheduling : Scheduling Queues, Schedulers, Context switch, Understanding basics of Process scheduling, Operations on Process – Process creation, Process termination, Understanding the system calls – fork(),wait(),exit(),Inter Process communication : Shared Memory, Message Passing, Pipe(),Understanding the need for IPC, PROCESS SYNCHRONIZATION: Background, Critical section Problem, Understanding the race conditions and the need for the Process synchronization																

UNIT:2**(12 Hours)**

PROCESS SYNCHRONIZATION: Peterson's solution, Synchronization Hardware, Understanding the two-process solution and the benefits of the synchronization hardware, Process synchronization: Semaphores, usage, implementation, Gaining the knowledge of the usage of the semaphores for the Mutual exclusion mechanisms, Classical Problems of synchronization – Readers writers problem, Bounded Buffer problem, Good understanding of synchronization mechanisms, Classical Problems of synchronization – Dining Philosophers problem (Monitor) Understanding the synchronization of limited resources among multiple processes, **CPU SCHEDULING:** FCFS,SJF,Priority, Understanding the scheduling techniques, CPU Scheduling: Round robin, Multilevel queue Scheduling, Multilevel feedback Scheduling, Understanding the scheduling techniques, Real Time scheduling: Rate Monotonic Scheduling and Deadline Scheduling, Understanding the real time scheduling, **DEADLOCKS:** Necessary conditions, Resource allocation graph, Deadlock prevention methods, Understanding the deadlock scenario, Deadlocks :Deadlock Avoidance, Detection and Recovery, Understanding the deadlock avoidance, detection and recovery mechanisms

UNIT:3**(14 Hours)**

MEMORY MANAGEMENT: Memory Management: Logical Vs Physical address space, Swapping, Understanding the basics of Memory management, Contiguous Memory allocation – Fixed and Dynamic partition, getting to know about Partition memory management and issues: Internal fragmentation and external fragmentation problems, Strategies for selecting free holes in Dynamic partition, Understanding the allocation strategies with examples, Paged memory management, Understanding the Paging technique. PMT hardware mechanism, Structure of Page Map Table, Understanding the components of PMT, Example : Intel 32 bit and 64 –bit Architectures, Understanding the Paging in the Intel architectures, Example : ARM Architectures, Understanding the Paging with respect to ARM, Segmented memory management, Understanding the users view of memory with respect to the primary memory, Paged segmentation Technique, Understanding the combined scheme for efficient management

UNIT:4**(14 Hours)**

VIRTUAL MEMORY– Background, Understanding the need of demand paging, **VIRTUAL MEMORY –** Basic concepts – page fault handling, Understanding, how an OS handles the page faults, Performance of Demand paging, Understanding the relationship of effective access time and the page fault rate, Copy-on write, Understanding the need for Copy-on write, Page replacement Mechanisms: FIFO, Optimal, LRU and LRU approximation Techniques, Understanding the Pros and cons of the page replacement techniques, Counting based page replacement and Page Buffering Algorithms, To know on additional Techniques available for page replacement strategies, Allocation of Frames - Global Vs Local Allocation, Understanding the root cause of the Thrashing, Thrashing, Causes of Thrashing, Understanding the Thrashing, Working set Model, Understanding the working set model for controlling the Working set Model.

UNIT:5**(14 Hours)**

STORAGE MANAGEMENT: Mass storage structure – Overview of Mass storage structure – Magnetic Disks, Understanding the Basics in storage management, Disk Scheduling, Understanding the various scheduling with respect to the disk, **FILE SYSTEM INTERFACE:** File concept, File access methods, Understanding the file basics, File sharing and Protection,

Emphasis the need for the file sharing and its protection, FILE SYSTEM IMPLEMENTATION: File system structure, To get the basic file system structure, Directory Implementation, Understanding the various levels of directory structure, FILE SYSTEM IMPLEMENTATION :Allocation methods, Understanding the pros and Cons of various disk allocation methods, FILE SYSTEM IMPLEMENTATION :Free space Management, Understanding the methods available for maintaining the free spaces in the disk, Swap space Management, Understanding the Low-level task of the OS.

Teaching Methods: Chalk& Board/ PPT/Video Lectures

Text Books/References:

1. *Operating System Principles*, Abraham Silberchatz, Peter B. Galvin, Greg Gagne 8th Edition, Wiley Student Edition.
2. *Operating systems - Internals and Design Principles*, W. Stallings, 6th Edition, Pearson.M. G. Say, “*Performance and design of AC machines*”, CBS Publishers, 2002.
3. *Modern Operating Systems*, Andrew S Tanenbaum 3rd Edition PHI.
4. *Operating Systems A concept - based Approach*, 2nd Edition, D. M. Dhamdhare, and TMH.
5. *Principles of Operating Systems*, B. L. Stuart, Cengage learning, India Edition.
6. *Operating Systems*, A. S. Godbole, 2nd Edition, TMH
7. *An Introduction to Operating Systems*, P.C.P. Bhatt, PHI.
8. *Operating Systems*, S, Haldar and A. A. Arvind, Pearson Education.
9. *Operating Systems*, R. Elmasri, A. G. Carrick and D. Levine, McGraw Hill.
10. *Operating Systems in depth*, T. W. Doeppner, Wiley.

Subject Code	Name of the Subject	L	T	P	C										
	Introduction to Soft Computing	3	0	0	3										
Course Educational Objectives															
CEO1	To introduce the concepts in Soft Computing														
CEO2	To introduce Fuzzy Logic based Systems														
CEO3	To introduce Neural Network based Systems														
CEO4	To introduce Genetic Algorithm based Systems and its hybrids														
Course Outcomes: Upon successful completion of this course, students should be able to:															
CO1	Understand the fundamental concepts of Soft Computing and differentiate between conventional and soft computing techniques														
CO2	Apply fuzzy logic concepts to solve real-world problems involving uncertainty and imprecision.														
CO3	Design and implement neural networks for pattern recognition and data analysis tasks.														
CO4	Use evolutionary algorithms for optimization and search problems in various domains.														
CO5	Integrate multiple soft computing techniques to develop hybrid systems for complex problem-solving.														
CO6	Analyze and evaluate the effectiveness of soft computing methods in various engineering and scientific applications.														
CO-PO & PSO Mapping															
COs	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	1	2	1												
CO2	2	2	2												
CO3	1	1	1												
CO4	1	3	1												
CO5	1	1	1												
CO6	1	1	3												
Avg.	1.1	1.6	1.5												
SYLLABUS															
UNIT:1 Basics of Soft Computing & Fuzzy Logic (12 Hours)															
Introduction to Soft Computing- Components of Soft Computing- Applications of Soft Computing - Fuzzy Sets – Basic Definition and Terminology – Set-theoretic Operations – Member Function Formulation and parameterization. Fuzzy Rules and Fuzzy Reasoning – Extension Principle and Fuzzy Relations – Fuzzy If-Then Rules –Fuzzy Reasoning – Fuzzy Inference Systems – Mamdani Fuzzy Models – Sugeno Fuzzy Models –Tsukamoto Fuzzy Models.															
UNIT:2 Neural Network (10 Hours)															
Concepts of Neural Network – Activation Functions – Architecture and characteristics of Neural Network - Supervised Learning Neural Networks – Perceptrons – Backpropagation Multilayer Perceptrons – ADALINE, MADALINE, Radial Basis Function Networks – Unsupervised Learning Neural Networks – Competitive Learning Networks – Kohonen Self-Organizing Networks .															

UNIT:3 Genetic Algorithm	(10 Hours)
Fundamentals of Genetic Algorithms – Creation of offspring – Encoding – Fitness function – Reproduction – Inheritance operators – Crossover – Inversion and deletion – Mutation - GA – Applications.	
UNIT:4 Neuro-Fuzzy Modelling	(10 Hours)
Adaptive Neuro-Fuzzy Inference Systems – Architecture – Hybrid Learning Algorithm – Learning Methods that Cross-fertilize ANFIS and RBFN.	
UNIT-5 Evolutionary Algorithms and Optimization Techniques	
Introduction to Evolutionary Algorithms: Genetic Algorithms, Genetic Programming, Operators in Evolutionary Algorithms: Selection, Crossover, Mutation, Hybrid Evolutionary Algorithms, Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO), Applications of Evolutionary Algorithms in Real-World Problems	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books:	
1. J.S.R.Jang, C.T.Sun and E.Mizutani, “Neuro-Fuzzy and Soft Computing”, Pearson, 2015. 2. Samir Roy, Udit Chakraborty, “Introduction to Soft Computing Neuro-Fuzzy and Genetic Algorithms”, Pearson, 2013	
Reference Books:	
1. Davis E.Goldberg, “Genetic Algorithms: Search, Optimization and Machine Learning”, Addison Wesley, 2009 2. Kumar S.Ray, “Soft Computing and Its Applications, Volume Two: Fuzzy Reasoning and Fuzzy Control”, CRC Press, 2014 3. Saroj Kaushik Sunita Tiwari, “ Soft Computing : Fundamentals Techniques & Applications”, Tata McGrawHill, 2018 4. Nunes Silva, I., Hernane Spatti, D., Andrade Flauzino, R., Liboni, L.H.B., dos Reis Alves, S.F., “Artificial Neural Networks - A Practical Course”, Springer, 2017 5. N.P.Padhy, S.P.Simon, “Soft Computing with MATLAB Programming”, Oxford University Press, 2015 6. Chakraverty, Snehashish, Sahoo, Deepti Moyi, Mahato, Nisha Rani, “Concepts of Soft Computing Fuzzy and ANN with Programming”, Springer, 2019. 7. Jamshid Ghaboussi, “Soft Computing in Engineering”, 1st Edition, CRC Press, 2018	

Subject Code		Name of the Subject								L	T	P	C		
		Fundamentals of Python Programming								2	0	2	3		
Course Educational Objectives															
CEO1	To understand the basics of programming using Python.														
CEO2	To Construct and execute basic programs in Python.														
CEO3	To implement database connectivity, CGI, Multithreading and web development in python.														
Course Outcomes: Upon successful completion of this course, students should be able to:															
CO1	Understand the basic concept of Python like data types and syntax of programming.														
CO2	Use python programming with user defined & built-in methods, objects of Python.														
CO3	Design application using the concepts of file, database access and exception handling.														
CO4	Create practical and contemporary applications such as web applications and discrete-event simulations, data analysis and IoT devices.														
CO5	Use regular expressions and oops concepts														
CO6	Design Database and implement Mutlithreading.														
CO-PO & PSO Mapping															
COs	PROGRAMME OUTCOMES														
	1	2	3	4	5	6	7	8	9	10	11	12			
CO1	3	1			1							1			
CO2	2	3	1	3	2							1			
CO3	2	3	3	3	2				1		1	2			
CO4	2	3	3	3	2				1		1	2			
CO5	1	2	2	1											
CO6	1	2	2	1											
Avg.	1.8	2.3	1.8	2.2	1.75				1		1	1.5			
SYLLABUS															
UNIT:1 (8 Hours)															
Introduction: Installation, First Python Program: Interactive Mode Programming, Script Mode Programming; Identifiers, Reserved Words, Lines and Indentation, Multi-Line Statements, Quotation & Comments; Assigning Values to Variables, Multiple Assignment.															
Standard Data Types: Numbers, Strings, Lists, Tuples, Dictionary; Data Type Conversion; Basic Operators: Arithmetic, Comparison, Assignment, Bitwise; Operators: Logical, Membership, Identity; Operators Precedence; Python Numbers & Mathematical functions, Python Strings.															
UNIT:2 (10 Hours)															
Python Program Flow Control : Conditional blocks using if, else and elif, Simple for loops in python, For loop using ranges, string, list and dictionaries, Use of while loops in python, Loop manipulation using pass, continue, break and else, Programming using Python conditional and loops block															
Python Functions, Modules And Packages:Organizing python codes using functions, Organizing python projects into modules, Importing own module as well as external modules,Understanding Packages, Powerful Lamda function in python, Programming using functions, modules and external packages															

Python String, List And Dictionary Manipulations : Building blocks of python programs, Understanding string in-built methods, List manipulation using in-built methods, Dictionary manipulation, Programming using string, list and dictionary in build functions	
UNIT:3	(12 Hours)
Python File Operation : Reading config files in python, Writing log files in python , Manipulating file pointer using seek, Programming using file operations, Understanding read functions, read(), readline() and readlines(), Understanding write functions, write() and writelines() Python Object Oriented Programming – OOPS: Concept of class, object and instances, Constructor, class attributes and destructors, Real time use of class in live projects, Inheritance , overlapping and overloading operators, Adding and retrieving dynamic attributes of classes , Programming using OOps support	
UNIT:4	(15 Hours)
Python Regular Expression : Powerful pattern matching and searching, Power of pattern searching using regex in python, Real time parsing of networking or system data using regex, Password, email, url validation using regular expression, Pattern finding programs using regular expression Python Exception Handling : Avoiding code break using exception handling, Safe guarding file operation using exception handling, Handling and helping developer with error code, Programming using Exception handling	
Unit-5	(10 Hours)
Python Database Interaction: SQL Database connection using python, Creating and searching tables, Reading and storing config information on database, Programming using database connections Python Multithreading: Understanding threads, forking threads, Synchronizing the threads, Programming using multithreading Python CGI Introduction : Writing python program for CGI application, Creating menus and accessing files, Server client program	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books: 1. “Python: The Complete Reference”, Martin C. Brown, McGraw-Hill/Osborne Media. 2. “Learning Python”, Mark Lutz, O’Reilly Media, Inc., Fifth Edition, 2013.	
Reference Books: 1. “Expert Python Programming”, Tarek Ziad, Packt Publishing. 2. “Python Essential Reference, David Beazley”, Dveloper's Library, Sams Publishing.	

Subject Code		Title of the subject (4 th Sem)									L	T	P	C	
		Advanced Communication Skills and Professional Ethics									1	0	0	1	
CourseEducationalObjectives															
CEO1	Todevelop communication skills andsoftskills ofthestudents.														
CEO2	Toenhanceemployabilityandentrepreneurialskills.														
CEO3	To inculcate professional ethics.														
Course Outcomes:The students will be able to:															
CO1	Develop ways to overcome barriers in professional communication														
CO2	Formulate advance speaking skills														
CO3	Organize professionals meetings and talks.														
CO4	Develop professional writing skills for career growth														
CO5	Understand the importance of ethics in professional life														
CO6	Use technological aids for self learning														
CO-POMapping															
COs	PROGRAMMEOUTCOMES														
	1	2	3	4	5	6	7	8	9	10	11	12			
CO1											3				
CO2										3					
CO3									3						
CO4										3					
CO5								3				2			
CO6										3					
SYLLABUS															
UNIT-1Advance Listening Skills (4hours)															
1.1 Barriers to effective listening at work place															
1.2 Overcoming the barriers.															
UNIT-2Advance Speaking Skill(3 hours)															
2.1 Barriers to the public speaking															
2.2 Overcoming the barriers															
UNIT-3 Advance Writing Skills (3hours)															
3.1 Barriers to professional writing.															
3.2 Overcoming the barriers.															
UNIT-4 Professional Ethics (4hours)															
4.1 What is Professional ethics, importance															

4.3 Citation; What, Why

5.1 What and why.

5.2 Spell check, Grammarly, Quill-bot, Ithenticate, Turnitin

Teaching Methods: Chalk&Board/PPT/Video Lectures

1. PracticalEnglishUsage.MichaelSwan,OUP,1995

2. Communication skills by Sanjay Kumar and Pushp Lata, Oxford University Press; Second Edition, 2015.

3. Advanced Communication English by Dr. O.V. Nachmuthu, The Cambridge Publishers; First Edition, 2011.

4. Professional Ethics in Engineering by Dr. C.Senthil Kumar, ARC Publications, 2017.

Subject Code		Name of the Subject (4 th Sem)								L	T	P	C		
		Advanced Communication Skills and Professional Ethics (Lab)								0	0	2	1		
Course Educational Objectives															
CEO1	To develop communication skills and soft skills of the students.														
CEO2	To enhance employability and entrepreneurial skills.														
CEO3	To inculcate professional ethics.														
Course Outcomes: The students will be able to:															
CO1	Develop ways to overcome barriers in professional communication														
CO2	Formulate advanced speaking skills														
CO3	Organise professional meetings and talks.														
CO4	Develop professional writing skills for career growth														
CO5	Understand the importance of ethics in professional life														
CO6	Use technological aids for self-learning														
CO-PO Mapping															
Cos	PROGRAMME OUTCOMES														
	1	2	3	4	5	6	7	8	9	10	11	12			
CO1											3				
CO2										3					
CO3									3						
CO4										3					
CO5								3				2			
CO6										3					
SYLLABUS															

Listening Scientific Documentaries	(2 hours)
Activity- Write down the important points in the documentaries	
Listening to a play/ short film	(2 hours)
Activity- summarise it in your own words	
Spontaneous Speaking	(2 hours)
Activity- Extempore	
Hosting a Meeting	(2 hours)
Activity- Hosting a mock meeting in a given topic	
Teach a class	(2 hours)
Activity-Giving lecture on any topic of choice to the class	
Give a Ted Talk	(2 hours)
Activity- Prepare and Deliver a Ted Talk	
Article Writing	(2 hours)
Activity- Analyze an article provided in class	
Social media post	(2 hours)
Activity- reading social media post for a company	
Learning the use of Job Portals	(2 hours)
Activity- Creating a profile Job Portal like linked in.	
Evaluating Each Other	(2 hours)
Activity-Using Pad let to create a social wall and evaluate each other's walls.	
Dialogue Writing	(2 hours)
Activity- Listening to a silent movie and writing dialogues/ plot.	
Skit Writing	(2 hours)
Prepare a skit and perform in front of the class	
Creative Writing	(2 hours)
Activity- Analyse a picture with a hidden theme and write a story based on it.	

Business Proposal (2 hours)
Activity- Techniques, types and practicing to write Business Proposal.
Role Play (2 hours)
Activity- Panel interview, task based mock interviews.
ReferenceBooks: 1. Communication skills by Sanjay Kumar and Pushp Lata, Oxford University Press; Second Edition, 2015. 2. Advanced Communication English by Dr. O.V. Nachmuthu, Cambridge Publishers; First Edition, 2011. 3. Advance English Communication Skills Lab by Lakshminarayanan, SCITECH Publications, 2009.

V Semester [Third Year]
B. Tech in Computer Science and Engineering (2023 Admission)

SN	CC	Sub Code	Name of the Subject	L	T	P	C	ISA	ESA
THEORY									
1	HS	23BMGMG35001 / 23BMGMG35002	Organizational Behaviour /Engineering Economics and Costing	2	0	0	2	40	60
2	PC	23BCSPC35001	Language Theory and Compiler Design	3	0	0	3	40	60
3	PC	23BCSPC35002	Data Mining and Data Warehousing	3	0	0	3	40	60
4	PE	23BCSPE350**	Professional Elective-III (From Bucket)	3	0	0	3	40	60
5	PC	23BCSPC35003	Software Engineering	3	0	0	3	40	60
6	OE	23BCSOE350**	Open Elective-I (From Bucket)	3	0	0	3	40	60
			TOTAL	17	0	0	17	240	360
PRACTICAL/SESSIONAL									
7	PC	23BCSPC35101	Language Theory and Compiler Design Lab	0	0	2	1	30	20
8	PC	23BCSPC35102	Data Mining and Data Warehousing Lab	0	0	2	1	30	20
9	PST	23BCSPW35101	Minor Project -I	0	0	4	2	50	50
10	PST	23BCSPW35102	Summer Internship-II	0	0	1	1	0	50
11	MC	23BMCMC35001	Constitution of India	0	0	0	0	#100	
			TOTAL	0	0	9	5	110	140
			TOTAL	17	0	9	22	350	500

Professional Elective Bucket

Sl. No.	Sem	Professional Elective III
1	23BCSPE35011	Cloud Computing
2	23BCSPE35012	Edge Computing
3	23BCSPE35013	Cloud Security

BUCKETS FOR OPEN ELECTIVES

Sl. No.	Sem	Open Elective I
1	23BCSOE35011	Web Technology
2	23BCSOE35012	Advanced Java
3	23BCSOE35013	Full Stack Development

Subject Code		Name of the Subject								L	T	P	C		
		Language Theory and Compiler Design								3	0	0	3		
Course Educational Objectives															
CEO1	To introduce the major concept areas of language translation and compiler design.														
CEO2	To enrich the knowledge in various phases of compiler and its use.														
CEO3	To provide practical programming skills necessary for constructing a compiler														
Course Outcomes: : <i>Upon successful completion of this course, students should be able to:</i>															
CO1	Understand the fundamental concepts and phases involved in the design and development of a compiler.														
CO2	Apply lexical and syntax analysis techniques for language translation using various parsing methods.														
CO3	Design and manage symbol tables and perform intermediate code generation for different programming constructs														
CO4	Implement semantic analysis and syntax-directed translation for type checking and expression evaluation.														
CO5	Develop code generation algorithms and optimize code for efficient memory and execution														
CO6	Use compiler generation tools to construct machine-dependent code and evaluate the effectiveness of various optimization techniques.														
CO-PO & PSO Mapping															
Cos	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	1	2	2												
CO2	2	1	2												
CO3	1	1	3												
CO4	3	2	3												
CO5	1	1	3												
CO6	1	1	1												
Avg.	1.5	1.3	2.3												
SYLLABUS															
UNIT:1 (12 Hours) Introduction to Compiler Design and Lexical Analysis, Overview of Compilation and its Phases, Lexical Analysis: Role of a Lexical Analyzer, Finite Automata: Non-Deterministic (NFA) and Deterministic (DFA), Regular Expressions, Lexical Grammar, and Design of a Lexical Analyzer, Tools for Lexical Analysis: Lex and Flex.															
UNIT:2 (12 Hours) Syntax Analysis: Role of a Parser, Context-Free Grammars, and Language s, Parse Trees and Derivations, Ambiguous Grammar, Top-Down Parsing: Recursive Descent Parsing, LL(1) Grammars, Bottom-Up Parsing: Shift-Reduce Parsing, SLR Parsers, LR(1) Parsers, LALR Parsers, Error Handling, Reporting, and Recovery Techniques															
UNIT:3 (10 Hours) Syntax-Directed Translation and Semantic Analysis, Syntax-Directed Definitions (SDD), Inherited and Synthesized Attributes, Dependency Graphs and Evaluation Orders, Semantic Analysis: Semantic Rules and Attribute Grammars, Applications of Syntax-Directed Translation															

UNIT:4 (10 Hours) Intermediate Code Generation and Symbol Table Management: Symbol Table: Structure, Attributes, and Scope Management, Intermediate Code Generation: Three Address Code, Quadruples, and Triples, Representations of Syntax Trees and DAG for Expressions, Type Checking, Type Conversions, and Control Flow Statements, Intermediate Code for Procedure Calls and Run-Time Environments
UNIT:5 Code Generation and Optimization, Code Generation: Basic Blocks, Flow Graphs, and Code Generation Techniques, Register Allocation and Assignment Techniques, Code Optimization: Peephole Optimization, Elimination of Local Common Sub-expressions, Flow Control Optimization, Removal of Redundant Code, and Unreachable Code, Introduction to Machine-Dependent Optimization
Teaching Methods: Chalk& Board/ PPT/Video Lectures/Lecture by Industry Expert
Text Books: 1. <i>Compilers – Principles, Techniques and Tools</i> Authors: Alfred V. Aho, Monica S. Lam, Ravi Sethi and Jeffrey D. Ullman Publisher: Pearson
Reference Books: 1. <i>Morden Compiler Design</i> , D. Galles, Pearson Education. 2. <i>Compiler Design in C</i> , Allen I. Holub, PHI

Subject Code		SUBJECT NAME											L	T	P	C
		Data Mining and Data Warehousing											3	0	1	4
Course Educational Objectives																
CEO 01	Understand the concepts of Association Rule Mining, Classification and Clustering algorithms.															
CEO 02	Applying various data mining techniques such as classification, clustering, association rule mining.															
CEO 03	Integrating data mining results into decision support systems and business intelligence applications to support decision-making.															
CEO 04	Apply best practices in data handling, ensuring data privacy and security.															
Course Outcomes: : Upon successful completion of this course, students should be able to:																
CO1	Understand and learn various Concepts and Techniques for gaining knowledge of fundamentals & techniques used for data mining and data warehouse.															
CO2	Learn various data mining techniques such as classification, clustering, association rule mining using tools.															
CO3	Develop skills in implementation of data mining algorithms on different datasets.															
CO4	Enhance the analytical and decision making abilities based on interesting patterns generated using data mining techniques.															
CO5	Gain knowledge on Outlier analysis techniques															
CO6	Understand the importance of applying Data mining concepts in different domains															
CO-PO & PSO Mapping (Articulation Matrix):																
COs		PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	01	02		
1	2	1	0	0	0	0	0	0	0	2	0	0				
2	2	3	1	0	1	0	0	0	0	1	0	0				
3	1	2	3	3	1	0	0	0	1	0	1	0				
4	1	1	2	2	1	0	0	0	1	0	1	1				
Avg.	1.5	1.75	1.5	1.25	0.75	0	0	0	0.5	0.75	0.5	0.25				
Justification of the Mapping:																
CO	PO	Level	Justification													
1	1	2	Understands the fundamental knowledge of statistics, mathematics, engineering basics on data mining and data warehouse.													
1	2	1	Analyses smaller problems on classification, clustering, association rule mining.													
1	10	2	Communicates using discussion on the different applications of data mining and warehouse in society.													
2	1	2	Allows to learn engineering fundamentals and mathematics to analyse the data, and understand the basic concepts.													
2	2	3	Enables students to identify, formulate, and analyze complex data problems													
2	3	1	Enables students to design solutions for complex problems using classification, clustering, and association rule mining.													
2	5	1	Learning the use of mining techniques and software tools for analysis and predictions.													
2	10	1	Helps students effectively communicate their result findings in group discussions, and design documents.													

3	1	1	It allows to understand basic principles and statistical knowledge during implementation of mining algorithms on different datasets.
3	2	2	It involves in analysing and demonstrating the problems using different datasets.
3	3	3	Developing skills for designing solutions for complicated problems for different application areas.
3	4	3	Conducting rigorous investigations, applying research-based knowledge, and experimenting to solve complex problems.
3	5	1	Involves in applying different modern tools, techniques for achieving desired results.
3	9	1	Engages in collaborative learning activities in a team or in individual while working with project problems
3	11	1	Allows a student to demonstrate about their project knowledge as team member or leader through presentations, reflecting foundational proficiency in communication.
4	1	1	Allows to give keen observations on basic concepts during implementations and pattern generations.
4	2	1	Analysing, formulating and reviewing the problem by observing data visualization
4	3	2	Designing solutions for complex problems by enhancing analytical and decision-making abilities through data mining techniques.
4	4	2	Learn investigating complex problems by using experimenting and research based on interesting patterns generated
4	5	1	Involves in utilizing modern tools, techniques, and resources, while actively engaged in analysis of the interesting patterns.
4	9	1	Allows to analyse the decision making patterns with group discussions in a team.
4	11	1	It enables to learn preparing models for decision making, demonstrate project knowledge with team effort, improves presentation skills.
4	12	1	It builds self-confidence and Curiosity to do more projects on predictions for different applications.

Syllabus

UNIT I		12 Hours
Introduction to Data Mining: Data Mining Technologies, Major issues in Data Mining, Data Objects and Attribute Types, Basic Statistical Descriptions of Data Data Preprocessing: Data Integration, Data Cleaning, Data Transformation, Data Reduction, Data Discretization and Data Summarization.		
UNIT II		12 Hours
Introduction to Data Warehousing: The Compelling need of Data Warehousing, Basic Concepts of Data Warehouse, Data Warehouse Architecture, Data Warehouse Models, ETL Operations, Meta Data in the Data Warehouse, Trends in Data Warehousing, Data Warehouse Modelling, OLAP Vs OLTP, Data Cube, Schemas for Multidimensional Data models, Data Warehouse Design and Usage.		
UNIT III		12 Hours
Introduction to Mining Frequent Pattern: Associations, Correlations: Market Basket Analysis, Apriori Algorithm, FP-Growth Algorithm, Mining Various kinds of association rules, from association rule mining to correlation analysis.		

Classification: Classification by Decision Tree Induction, Bayes Classification Methods, Rule Based Classification, KNN Classifier, Logistic Regression, Support Vector Machine.		
UNIT IV		12 Hours
Introduction to Cluster Analysis: Types of Data in Cluster Analysis Categorization of major clustering methods: K-means Clustering, K-medoid Clustering, Agglomerative Clustering, Hierarchical Clustering, DBSCAN Clustering. Clustering Graph and Network Data, Anomaly Detection: Outlier Analysis, Statistical Analysis, Distance Based Analysis, Clustering Based Analysis, Text Mining: Text Preprocessing, IF-IDF, Text Classification, Text Clustering, Web mining: Web content mining, Web structure mining, Web usage mining, Applications of Data Mining in different area.		
UNIT-V		[8 hours]
Challenges of outlier detection, Outlier detection methods: Introduction, Supervized and Semi-supervised methods, Unsupervised methods, Statistical and Proximity based methods, Statistical approaches, Statistical data mining.		
Teaching Methods: Chalk& Board/ PPT/Video Lectures/Lecture by Industry Expert/MOOCs		
Text Books: 3. Data Mining: Concepts and techniques: J. Han, M. Kamber, J. Pei, Elsevier 4. Data Warehousing Fundamentals: Paulraj Ponniah, Willey India.		
Reference Book: 1. Data Mining –A Tutorial based primer by R.J.Roiger, M.W.Geatz, Pearson Education. 2. Data Mining & Data Warehousing Using OLAP: Berson, TMH. 3. Data Warehousing: Reema Thareja, Oxford University Press		

Subject Code		SUBJECT NAME										L	T	P	C
		Cloud Computing										3	0	0	3
Course Educational Objectives															
CEO 01	Understand the rationale behind the cloud computing revolution														
CEO 02	Introduce various models of cloud computing														
CEO 03	Understand how to design applications on cloud and the role of security														
CEO 04	Understand and design distributed systems for scalability														
Course Outcomes: Upon successful completion of this course, students should be able to:															
CO1	Understand the technical and business rationale behind cloud computing														
CO2	Outline the model of cloud computing to use for solving a particular problem														
CO3	Design and Build applications for the cloud and understand the security implications														
CO4	Understand and Apply the fundamentals of distributed systems design to cloud computing														
CO5	Design and develop cloud-based service applications using appropriate platforms and tools.														
CO6	Explore and assess emerging trends in cloud computing for future-proofing cloud solutions.														
CO-PO & PSO Mapping s(Articulation Matrix):															
COs		PROGRAMME OUTCOMES											PSOs		
	01	02	03	04	05	06	07	08	09	10	11	12	01	02	
01	3	1	1												
02	3	2	1												
03	3	2	0												
04	3	3	2												
Avg.	3	2	1												
Justification of the Mapping:															
CO	PO	Level	Justification												
1	1	3	It has the highest significance												
1	2	1	It has the least significance												
1	3	1	It has the least significance												
2	1	3	It has the highest significance												
2	2	2	It has an average significance												
2	3	1	It has the least significance												
3	1	3	It has the highest significance												
3	2	2	It has an average significance												
4	1	3	It has the highest significance												
4	2	3	It has the highest significance												
4	3	2	It has an average significance												
Syllabus															

UNIT I	12 Hours
Cloud Computing: Introduction, Types of cloud: private, public and hybrid cloud. Hardware and Infrastructure: IaaS, PaaS, SaaS. public vs private clouds, Benefits and challenges of cloud computing, Virtualization : Types of Virtualization, Implementation Levels , Virtualization Structures , Tools and Mechanisms, Virtualization of CPU, Memory, I/O Devices ,Virtual Clusters and Resource management – Virtualization for Data-center Automation.	
UNIT II	14 Hours
Cloud infrastructure: Architectural Design of Compute and Storage Clouds, Layered Cloud Architecture Development, Design Challenges, Inter Cloud Resource Management, Resource Provisioning and Platform Deployment, Global Exchange of Cloud Resources. Cloud Applications: Technologies and the processes required when deploying web services; Deploying a web service from inside and outside a cloud architecture, advantages and disadvantages Benefits and challenges to Cloud architecture, Application availability, performance, security and disaster recovery; SLA Management: Types of SLA, Life Cycle of SLA, SLA Management	
UNIT III	10 Hours
Cloud Economics: Economics of choosing a Cloud platform for an organization, based on application requirements, economic constraints and business needs (e.g. Amazon, Microsoft and Google, Salesforce.com, Ubuntu and Redhat); Cloud Simulator: Benefits of CloudSim, Architecture of Cloud Sim, Installation procedures, Types of Scheduling Algorithms used in Cloud resource Management	
UNIT IV	10 Hours
Cloud security: Security Overview, Cloud Security Challenges and Risks, Software-as-a-Service Security, Security Governance, Risk Management, Security Monitoring, Security Architecture Design, Data Security, Application Security, Virtual Machine Security, Identity Management and Access Control, Autonomic Security. Cloud based service, applications and development platform.	
UNIT V	10 Hours
Cloud Service Development and Emerging Trends: Cloud-based Service Applications, Platforms and Tools for Cloud Development, Development and Deployment of Cloud Services, Emerging Trends in Cloud Computing: Edge Computing, Serverless Computing, Containerization. Case Studies: Real-world Cloud Implementations and Best Practices, Future Directions: Quantum Cloud Computing and AI in the Cloud.	
Teaching Methods: Chalk& Board/ PPT/Video Lectures/Lecture by Industry Expert/MOOCs	
Text Books: 1. Cloud Computing : A Practical Approach by Anthony T. Velte Toby J. Velte, Robert Elsenpeter, 2010 by The McGraw-Hill 2. Gautam Shroff, “Enterprise Cloud Computing Technology Architecture Applications”, Cambridge University Press; 1 edition, [ISBN: 978-0521137355],2010 3. Rajkumar Buyya, Christian Vecchiola, S.Tamarai Selvi, ‘Mastering Cloud Computing’, TMGH, 2013.	

Subject Code		Name of the Subject								L	T	P	C			
		Software Engineering								3	0	0	3			
Course Educational Objectives																
Course Outcomes: <i>Upon successful completion of this course, students should be able to:</i>																
CO1	Learn various software development process models and their suitability.															
CO2	Able to apply the methods of requirement elicitation.															
CO3	Learn Conventional Software Management and Evolution of Software Economics.															
CO4	Explain how a software process flows and understand Project Organizations and Responsibilities.															
CO5	Apply Agile methodologies and DevOps practices to enhance software development efficiency and collaboration															
CO6	Explore and integrate emerging trends and advanced techniques in software engineering for future-proof software solutions.															
CO-PO & PSO Mapping																
Cos	PROGRAMMEOUTCOMES												PSOs			
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	
CO1		3				2				1	1					
CO2		2		3												
CO3	1		1		2											
CO4		1														
Avg.	1	2	1	3	2	2				1	1					
SYLLABUS																
UNIT:1 (10 Hours)																
Introduction to Software Engineering, Software Process, Perspective and Specialized Process Models: The Waterfall Model, Incremental Process Models, The RAD Model, Prototyping Model, V- Model, Spiral Model, Agile and Scrum Model. COCOMO Model																
UNIT:2 (12 Hours)																
Software Requirements: Functional and Non-functional, User requirements, System requirements, Software Requirements Document, Requirement Engineering Process: Feasibility Studies, Requirement elicitation and analysis, Requirement validation, Requirements Management. CASE TOOL: Application on Documentation																
UNIT:3 (12 Hours)																
Design process: Design concepts, Design Model, Design Heuristic Architectural Design: Architectural Mapping using Data Flow Structure Chart design using DFD. Transform Analysis, Transaction Analysis: OOD Modeling Using UML, User Interface Design.																
UNIT:4 (14 Hours)																
Software implementation techniques: coding practices: Refactoring, Code Review, Code Inspection, Driver and Stub Module. Software Testing Fundamentals: Internal and external views of Testing- White Box Testing, Basis path testing, control structure testing. Black Box Testing: Regression Testing, Unit Testing, Integration Testing, Validation Testing, System Testing and Debugging. CASE TOOL: Test-Case Design, Software Reliability Measures and Growth Modeling: SEI CMM: Characteristics of Software Maintenance, Reverse Engineering, Re-engineering.)																
UNIT-V																
Emerging Trends and Advanced Topics in Software Engineering, Agile Software Development: Scrum, Kanban, Extreme Programming (XP), and Lean DevOps: Principles, Practices, and Tools , Cloud-Based																

Software Engineering and Microservices Architecture, Secure Software Development: Secure Coding Practices, Threat Modeling, and Vulnerability Analysis, Future Trends: AI in Software Engineering, Quantum Computing, and Green Software Engineering

Teaching Methods: Chalk& Board/ PPT/Video Lectures

Text Book:

1. *Fundamentals of Software Engineering*, Rajib Mall, PHI, 2014.
2. *Walker Rayce : Software Project Management*, Pearson Education, 2005

Reference Book

1. *Software Engineering, A Practitioner's Approach*, Roger S. Pressman, TMG Hill.
2. *Software Engineering*, I. Sommerville, 9th Ed. , Pearson Education
3. *Richard H.Thayer: Software Engineering Project Management*, IEEE Computer Society, 1997
4. *ShereK.D. : Software Engineering and Management*, Prentice Hall, 1988

SUBJECT CODE		SUBJECT NAME						L	T	P	C			
		Advanced Java Programming						3	0	0	3			
Course Educational Objectives														
CEO1	Objective of this course is to provide the ability to design console based, GUI based and web based applications. Students will also be able to understand integrated development environment to create, debug and run multi-tier and enterprise-level applications.													
Course Outcomes: : <i>Upon successful completion of this course, students should be able to:</i>														
CO1	design and develop data driven web applications using JDBC, servlet and JSP API.													
CO2	develop MVC based web applications using SpringBoot.													
CO3	understand significance of Object Relational Mapping (ORM) in Java context.													
CO4	construct ORM-based web applications using Hibernate.													
CO5														
CO6														
CO-PO & PSO Mapping (Articulation Matrix):														
Cos	PROGRAMME OUTCOMES												PSOs	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	1	1	3				3						
CO2		3	2					3						
CO3	2													
CO4		2	2											
Avg.	2.5	2	1.6					3						
SYLLABUS														
UNIT:1 (12 Hours) Basics of JDBC:Introduction to JDBC, Need of JDBC, JDBC Drivers (4 types), Architecture of JDBC, Components of JDBC (Classes and Interfaces). Programming with JDBC: Creating a DATABASE (ORACLE) , First Program to connect to the DATABASE created, Loading the Driver, Establishing the Connection, Creating Statements (Statement/ PreparedStatement/ CallableStatement), Executing a SQL Query, Different types of SQL Queries. JDBC Program to retrieve data from DATABASE: Introduction to Result Set, Result Set with Statement Interface, Result Set with PreparedStatement Interface, Updating and deleting data in database, ResultSetMetadata, Executing Stored Procedures Using Callable Statement.														
UNIT:2 (12 Hours) Web Architecture: Introduction to HTML, Describe the various HTML tags and their usage. Layout Tags, Semantic Tags, HTML table, forms, style, div. Introduction to Servlets: What is Servlet, Advantage of Servlet Over Applets and CGI, Strengths of Servlet, Architecture of Web Application, Web Servers and its Containers, Role of servlet in Web application development, Understanding servlet-api, Understating HTTP protocol and communication between HTML-SERVLET. Getting Deep to Servlets: Types of Servlet, Difference between HttpServlet and GenericServlet, Life cycle of Servlets and different life cycle methods, Difference between doGet() and doPost(), Servlet Generating Html output, Collecting Client submitted data in a Servlet.														

Servlet communications: Servlet to DBMS communication using type-4 connection, Servlet to DBMC communication using JDBC connection pooling, Servlet communication with other servlets (Servlet Chaining), Servlet communication with JSP or HTML page (sendRedirect(), Difference between sendRedirect() and RequestDispatcher forward(), Understanding ServletConfig. Conclusion to Servlets: Servlet Filters and wrappers, Servlet Listeners, Session Tracking, Cookies, HttpSession, HTML hidden form filed element, URL rewriting, Annotation based servlet programs.	
UNIT:3	(10 Hours)
Java Server Pages:Introduction to JSP, Scope of JSP, Anatomy of a JSP program, xecution of a JSP program, Significance of JSP Engine, Built in objects of JSP, Significance of JSP Elements, Scripting Elements, Scriptlets, Declaration, Expression, Directives and Action Elements, Page Directive, Include Directive, Taglib Directive, Forward action element, Include, Param, useBean with introduction to beans, setProperty, getProperty	
UNIT:4	(14 Hours)
Hibernate : Introduction to Hibernate, Architecture of hibernate, Hibernate O/R mapping, Configuring hibernate development environment, Implementing hibernate O/R Mapping, Introduction to Hibernate Query Language (HQL)	
UNIT-5	
Sping and SpringBoot : Spring Basic, Spring IOC, Spring MVC, Spring MVC with Hibernate. Introduction to SpringBoot, SpringBoot Injection, dependency, autowired, SpringBoot JPA, SpringBoot Webapp MVC.	
Teaching Methods: Chalk& Board/ PPT/Video Lectures/Lecture by Industry Expert/MOOCs	
Text Books: 1. Advanced Java Programming, Uttam K. Roy, Oxford University Press.	
Reference Book: 1. Black book, Kogent Learning Solution Inc. 2. Java 2: The Complete Reference by Herbert Schildt, Fifth Edition Paperback	

VI Semester [Third Year]
B. Tech in Computer Science and Engineering (2023 Admission)

SN	CC	Sub Code	Name of the Subject	L	T	P	C	ISA	ESA
THEORY									
1	HS	23BMGMG35001 / 23BMGMG35002	Engineering Economics and Costing/ Optimization Engineering	2	0	0	2	30	70
2	PC	23BCSPC36001	Introduction to Machine Learning	3	0	0	3	30	70
3	PC	23BCSPC36002	Microprocessors and Microcontroller	3	0	0	3	30	70
4	PE	23BCSPE360**	Professional Elective-IV (From Bucket)	3	0	0	3	30	70
5	PC	23BCSPC36003	Introduction to Internet of Things	3	0	0	3	30	70
6	OE	23BCSOE360**	Open Elective-II (From Bucket)	3	0	0	3	30	70
		23BMCMC36001	TOTAL	17	0	0	17	180	420
PRACTICAL/SESSIONAL									
7	PC	23BCSPC36103	Introduction to Internet of Things Lab	0	0	2	1	30	20
8	PC	23BCSPC36102	Microprocessors and Microcontrollers Lab	0	0	2	1	30	20
9	PST	23BCSPW36101	Minor Project Work-II	0	0	4	2	50	50
			TOTAL	0	0	8	4	110	90
			TOTAL	17	0	8	21	290	510

Professional Elective Bucket

Sl. No.	Sem	Professional Elective IV
1	23BCSPE36011	Digital Image Processing
2	23BCSPE36012	Pattern Recognition
3	23BCSPE36013	Introduction to Digital Signal Processing

BUCKETS FOR OPEN ELECTIVES

Sl. No.	Sem	Open Elective II
1	23BCSOE36011	Android Application Development using Kotlin
2	23BCSOE36012	Mobile Computing
3	23BCSOE36013	IOS Development

Assignment Based Assessment

Subject Code		Name of the Subject								L	T	P	C			
		Introduction to Machine Learning								3	0	0	3			
Course Educational Objectives																
CEO1: Introduce Basics of Matrices, Random Variables and Distributions relevant for the study of Machine Learning Techniques																
CEO2: Formulate a well defined machine learning problem with clear Metrics																
CEO3: Familiarize with techniques for Dimensionality reduction and Computational Efficiency																
CEO4: Understand the notions of Hypotheses Space, Hypotheses Structure and Search																
Course Outcomes: : Upon successful completion of this course, students should be able to:																
CO1	Identify and classify different categories of Data attributes, Dimensions, Sample sizes															
CO2	Understand and Apply Supervised, Unsupervised Learning techniques															
CO3	Discriminate classifications based on Logistic and Linear Regression and Function Estimation															
CO4	Produce Rules and Associations for impactful recommendations from data															
CO5	Apply Neural Networks and design the algorithms															
CO6	Understand the applications of ML in various areas															
CO-PO & PSO Mapping																
Cos	PROGRAMME OUTCOMES												PSOs			
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	
CO1	3	2														
CO2	3	2	2													
CO3	3	3														
CO4	2	3														
Avg.	2.75	2.5	2													
SYLLABUS																
UNIT:1 (12 Hours)																
Introduction to Machine Learning																
Introduction ,Components of Learning , Learning Models , Geometric Models, Probabilistic Models, Logic Models, Grouping and Grading, Designing a Learning System, Types of Learning, Supervised, Unsupervised, Reinforcement, Perspectives and Issues, Version Spaces, PAC Learning, VC Dimension. Understand your Data and Prepare Your Data For Analysis, Exploratory Data Analysis, Basic probability theory, Random variables, Probability distributions and Sampling Distributions, Markov models, Bayesian learning, Applications.																
UNIT:2 (10Hours)																
Supervised and Unsupervised Learning																
Decision Trees: ID3, Classification and Regression Trees, Regression: Linear Regression, Multiple Linear Regression, Logistic Regression,																
Support Vector Machines: Linear and Non-Linear, Kernel Functions, K Nearest Neighbors. Introduction to clustering, K-means clustering, K-Mode Clustering.																
Feature Engineering: Dimensionality reduction: PCA, LDA Feature Scaling, and Feature Selection: Filter methods, Wrapper methods, Embedded methods.																

UNIT:3	(12 Hours)
Ensemble and Probabilistic Learning Model	
Combination Schemes, Voting, Error-Correcting Output Codes, Bagging: Random Forest Trees, Boosting: Adaboost, Stacking. Gaussian mixture models - The Expectation-Maximization (EM) Algorithm, Information Criteria, Nearest neighbour methods - Nearest Neighbour Smoothing, Efficient Distance Computations: the KD-Tree, Distance Measures.	
UNIT:4	(14Hours)
Neural Networks	
Introduction, Early Models, Perceptron Learning, Backpropagation, Initialization, Training & Validation, Parameter Estimation - MLE, MAP, Bayesian Estimation Unsupervised learning, Partitional Clustering, Hierarchical Clustering, Density-based Clustering, Gaussian Mixture Models, Expectation Maximization. Introduction to reinforcement and explanation-based learning. ML applications in Computer Vision, Recommendation Systems: Case Study Approach	
Unit-5	(10 Hours)
Advanced Topics and Applications in Machine Learning: Reinforcement Learning: Markov Decision Process (MDP), Q-Learning, Deep Q-Networks (DQN), Natural Language Processing (NLP): Text Processing, Word Embeddings, and Transformers, Generative Models: Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs), Model Evaluation and Hyperparameter Tuning: Cross-Validation, Grid Search, Random Search, Future Trends: Explainable AI, Federated Learning, and Quantum Machine Learning	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books:	
1. Machine Learning, <i>Tom M. Mitchell,(2013), McGraw-Hill Education (INDIAN EDITION)</i> 2. Hands-On Machine Learning with Scikit-Learn and TensorFlow, <i>AurélienGéron, (2017) O'Reilly</i> 3. Introduction to Machine Learning with Python A guide for data scientists, <i>Andreas, C. Muller & SarahGuido,(2016)O'Reilly</i>	
Reference Books:	
1. <i>EthemAlpaydin, (2013), Introduction to Machine Learning, 2nd Ed., MIT Press.</i>	

Subject Code		Name of the Subject								L	T	P	C			
		Android App Development Using Kotlin								4	0	1	4			
Course Educational Objectives																
CEO1	This course covers the fundamentals of Kotlin Programming and Android programming using the Android SDK. Topics discussed in this course include: fundamental concepts in Android programming - activities and intents, designing user interface using views, data persistence, content providers, messaging and networking, location-based services, and developing android services.															
Course Outcomes: Upon successful completion of this course, students should be able to:																
CO1	demonstrate their understanding of Kotlin Programming.															
CO2	demonstrate their understanding of the fundamentals of Android operating systems, development trends and Android platform.															
CO3	Design kotlin based stand android application.															
CO4	Develop android application using API and database.															
CO5	Implement different types of Menus															
CO6	Usage of Data Storage and SOLite															
CO-PO & PSO Mapping																
COs	PROGRAMME OUTCOMES															
	1	2	3	4	5	6	7	8	9	10	11	12				
CO1	3	1			2							1				
CO2	2	3	1	3	2							1				
CO3	2	3	3	3	2				1		1	2				
CO4	2	3	3	3	2				1		1	2				
Avg.	2.25	2.5	2.3	3	2				1		1	1.5				
SYLLABUS																
UNIT:1 (8 Hours)																
Introduction to Kotlin: Kotlin Architecture, Basic syntax, Kotlin Comments, Kotlin Keywords, Defining variable, Use of variables, Data types- (Booleans, Strings, Arrays, Ranges), Operators, Control Flow, if... else expression, for loops, While loops, Break and Continue, functions in Kotlin, user defining functions. Kotlin Collections (Lists, Sets, Maps), Kotlin objects and Classes, Constructors, Inheritance, Interface, Visibility Control, Extension, Data Classes, Sealed Class, Generics, Delegation, Destruction, Exception Handling.																
UNIT:2 (15 Hours)																
Introduction to Android: Introduction to Android Operating System, Android Architecture, Core Building Blocks, Android Emulator, Application Structure, Setting up development environment, Dalvik Virtual Machine & .apk file extension, AndroidManifest.xml, uses-permission & uses-sdk, Resources & R.java o Assets, Layouts & Drawable Resources, Fundamentals: Basic Building blocks - Activities, Services, Broadcast, Receivers & Content providers, UI Components - Views & notifications, Components for communication -Intents & Intent Filters Android API levels (versions & version names), User Interface Architecture: Application context, intents, Activity life cycle, Services, Android widgets, UI widgets, Working with Buttons, Toast, Custom Toast, Toggle Button, Check Button, Custom Check Button, Radio Button, Dynamic Radio button, Alert Dialog, Android Views: Text view, Autocomplete Text view, List View, Custom List View, Rating Bar, WebView Seek Bar, Date Picker, Time Picker, Analog and Digital, Progress Bar, Android Layouts: Linear Layout, Relative Layout, Table Layout, Table Layout with Frame Layout, Search View, Search View on Toolbar, Edit Text with Text Watcher.																

UNIT:3	(10 Hours)
Switchers: Switch, Dynamic Switch, Text Switcher, Dynamic Text Switcher, Image Switcher, Dynamic Image Switcher. Spinner: Spinner, Dynamic Spinner, Dynamic TextClock, Chronometer, Dynamic Chronometer, Notifications, Android Toast, Dynamic RadioGroup, Android Slide Up/Down in Kotlin, Scroller: Vertical Scroll, View Horizontal Scroll View, Image Switcher, Image Slider, View Stub, XML Parsing using DOM Parser, XML Parsing using SAX Parser, Kotlin Android XMLPullParser	
UNIT:4	(10 Hours)
Menu : Option menu, Context menu, Sub menu, menu from xml, menu via code, Examples. Navigation: Create a fragment, Define navigation paths, Start an external activity, Kotlin Regex: Regular Expressions, Kotlin Annotations, Adapters, Android RecyclerView, Testing Android applications, Publishing Android application, Using Android preferences, Managing Application resources in a hierarchy, working with different types of resources	
UNIT:5	
Data Storage: Shared Preferences, Android File System, Internal storage, External storage, SQLite Firebase. Common Android APIs: Using Android Data and Storage APIs etc.	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books:	
1. Bruce Eckel and Svetlana Isakova (2021), “Atomic Kotlin”, Mindview LLC, Crested Butte, CO 2. WeiMeng Lee (2012), “Beginning Android Application Development”, Wrox Publications (John Wiley, New York) (For 1 to 5 units).	
Reference Books:	
1. Ed Burnette (2010), “Hello Android: Introducing Google's Mobile Development Platform”, The Pragmatic Publishers, 3rd edition, North Carolina USA 2. Reto Meier (2012), “Professional Android 4 Application Development”, Wrox Publications (John Wiley, New York). 3. ZigurdMednieks, Laird Dornin, Blake Meike G, Masumi Nakamura (2011), “Programming Android: Java Programming for the New Generation of Mobile Devices”, O'Reilly Media, USA	
e-Contents:	
1. https://developer.android.com/kotlin 2. https://developer.android.com/studio	

Subject Code		Name of the Subject						L	T	P	C				
		Microprocessors and Microcontrollers						3	0	2	4				
Course Educational Objectives															
CEO1	To Develop assembly language programs and basic concepts of the microprocessor and microcontroller														
CEO2	To provide solid foundation on interfacing the external devices to the microprocessor & microcontroller according to the user requirements in order to create novel products and solutions for the real time problems														
CEO3	To Familiar and Design of any type of embedded systems related to industrial and real time applications by knowing the concepts of Microprocessor and Microcontrollers.														
CEO4	To assist the students with an academic environment aware of excellence guidelines and lifelong learning needed for a successful professional carrier in the field embedded systems.														
Course Outcomes: Upon successful completion of this course, students should be able to:															
CO1	Understand the architecture of 8085, 8086 & 8051														
CO2	Understand and analyze different data transfer schemes and its applications														
CO3	To design circuits and Implement programs using microcontrollers.														
CO4	Understand the need & use of Interrupt structure in 8085, 8086 & 8051.														
CO5	Analyze and troubleshoot microprocessor and microcontroller systems.														
CO6	Apply microcontrollers in real-world embedded system applications.														
CO-PO & PSO Mapping															
Cos	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	3	1													
CO2	3	2													
CO3	2	2	1												
CO4	3	2													
CO5	1	1													
CO6	2	2													
Avg.	2.3	1.6	1												
SYLLABUS															
UNIT:I (10 Hours)															
Introduction to 8 bit Microprocessors															
Introduction to 8085 microprocessor, Architecture, Signal Descriptions ,Buses-Address bus ,data bus and control Bus ,Instruction format ,Instruction sets ,addressing Modes, Assembly Language Programming ,Timing diagram, stack and sub routine, Data Transfer Schemes ,Memory Interfacing and 8085 interrupts.															
UNIT:2 (12 Hours)															
Advanced Microprocessor															
Introduction to 8086 microprocessor, 8086 Architecture, Register Organization, signal descriptions, Memory Segmentation. Physical memory organization. Addressing Modes, instruction Set .Minimum and Maximum mode operation, Bus Cycle of minimum mode and maximum mode. Interrupts of 8086, Memory interfacing & Assembly Language Program.															

UNIT:3 Peripheral Devices (10 Hours) Programmable Peripheral Interface (8255), Programmable Interval Timer (8254) Programmable Interrupt Controller (8259A) - Programmable DMA Controller(8257),Programmable Communication Interface (8251A) – Programmable Keyboard and Display Controller (8279).
UNIT:4 (12 Hours) 8051 Microcontroller Overview of 8051 microcontroller. Architecture. I/O Ports. Memory organization, Addressing modes, data transfer instructions, Logical instructions, Arithmetic instructions, Branching (Jump & Call) instructions, Bit addressable instructions and special instructions, Interrupts and interrupt handler sub routines (Interrupt Service Routines).Assembly language program.
UNIT-5 (10 Hours) Advanced Microcontrollers and Embedded Systems: Introduction to advanced microcontrollers: ARM, AVR, and PIC, Comparison of different microcontroller architectures, Embedded system design: Hardware/software co-design, RTOS basics, Application of microcontrollers in embedded systems, Case studies: Real-time applications using microcontrollers
Teaching Methods: Chalk& Board/ PPT/Video Lectures
Text Books: 1. <i>Microprocessor Architecture, Programming and application with 8085</i> , R.S. Gaonkar, PRI Penram International publishing PVT. Ltd., 5thEdition 2. <i>Advanced Microprocessors and Peripherals</i> - A. K. Ray and K.M. Bhurchandani, TMH, 2nd edition 2006. 3. <i>The 8051 Microcontroller and Embedded Systems</i> , Muhammad Ali Mazidi, Janice Gillispie Mazidi, Rolin D.M C Kinlay, Pearson Education, Second Edition, 2008.
Reference Books: 1. <i>Microprocessors and Microcontrollers Architecture, programming and system design using 8085, 8086, 8051 and 8096</i> , Krishna Kant, PHI Publication, 2007. 2. <i>Microprocessors and Interfacing, Programming and Hardware</i> , Douglas V Hall, TMH Publication, 2006. 3. <i>Microprocessors and Interfacing</i> , N. Senthil Kumar, M. Saravanan, S. Jeevananthan and S.K. Shah, Oxford University Press.

Subject Code	Name of the Subject										L	T	P	C	
	Introduction to Digital Image Processing										3	0	0	3	
Course Educational Objectives															
CEO1: To provide deep understanding of basic concepts of digital image acquisition															
CEO2: T provide deep Understanding of various digital image enhancement techniques															
CEO3: Understand image restoration and segmentation methods															
CEO4: To provide understanding and implementation of image compression techniques															
Course Outcomes: : <i>Upon successful completion of this course, students should be able to:</i>															
CO1	Understand basics of digital images and tools for image processing														
CO2	Learn and implement image Enhancement techniques														
CO3	Understand and Learn image Restoration and Segmentation Methods														
CO4	Understand and implement Image Compression techniques														
CO5	Evaluate the performance of image processing techniques in various applications.														
CO6	Utilize software tools and programming languages to solve practical image processing problems.														
CO-PO & PSO Mapping															
Cos	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	3	2													
CO2	3	2	2												
CO3	3	3													
CO4	3	3													
CO5															
CO6															
Avg.	3	2.5	2												
SYLLABUS															
UNIT:1 (14 Hours)															
Introduction, Origin- Steps in Digital Image Processing, Components, Elements of Visual Perception, Image Sensing and Acquisition, Image Sampling and Quantization, Relationships between pixels, Introduction to Image processing toolbox in MATLAB, Tool box practice, Exploring functions, Introduction to Spatial Domain, Gray level transformations, Histogram processing, Basics of Spatial Filtering, Smoothing and Sharpening Spatial Filtering, Frequency Domain: Basics of filtering, Smoothing and Sharpening frequency domain filters, Smoothing and Sharpening frequency domain filters, MATLAB code for histogram equalization, MATLAB code for spatial and frequency domain filter.															
UNIT:2 (12 Hours)															
Noise models – Mean Filters – Order Statistics, Adaptive filters – Band reject Filters – Band pass Filters, Inverse Filtering – Wiener filtering Segmentation, Point, Line, and Edge Detection, Marr-Hildreth & Canny edge detector, Edge Linking and Boundary detection, Local & Regional processing-Region based segmentation, Morphological processing- Watershed segmentation algorithm, MATLAB code for restoring an image after degradation using adaptive and wiener filter, Edge detection operator															

UNIT:3	(14 Hours)
Wavelets – Subband coding – Multiresolution expansions, Fundamentals of Compression – Image Compression methods - Error Free Compression, Variable Length Coding – Bit-Plane Coding – Lossless Predictive Coding, Lossy Compression – Lossy Predictive Coding, Compression Standards- Huffman, Arithmetic coding, LZW coding, Run Length Encoding, Compression Standards, Huffman, Arithmetic coding, LZW coding, Run Length Encoding, Block Transform coding, Wavelet coding, JPEG standard, MATLAB code for image compression: Huffam coding, Arithmetic coding, wavelet coding.	
UNIT:4	(10 Hours)
Boundary representation – Chain Code, Polygonal approximation, signature, boundary segments, Boundary description – Shape number Fourier Descriptor, Regional Descriptor, Topological - Texture – Patterns and Pattern classes, Recognition based on matching, MATLAB code for image representation, MATLAB code for image recognition, MATLAB practice exercise.	
UNIT-V	(10 Hours)
Applications and Advanced Topics	
Introduction to advanced topics: machine learning in image processing, deep learning for image analysis, Applications in medical imaging, remote sensing, and computer vision, Case studies of image processing in real-world scenarios, Overview of current trends and future directions in digital image processing, Course project: Application of image processing techniques to a specific problem	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books:	
1. Rafael C. Gonzales, Richard E. Woods, “Digital Image Processing”, Third Edition, Pearson Education, 2010.	
Reference Books:	
1. Rafael C. Gonzalez, Richard E. Woods, Steven L. Eddins, “Digital Image Processing Using MATLAB”, Third Edition Tata Mc Graw Hill Pvt. Ltd., 2011.	
2. Anil Jain K. “Fundamentals of Digital Image Processing”, PHI Learning Pvt. Ltd., 2011.	
3. William K Pratt, “Digital Image Processing”, John Willey, 2002.	

Subject Code		Name of the Subject								L	T	P	C	
		Introduction to Internet of Things								3	0	2	4	
Course Educational Objectives														
CEO1	The objective of this course is to impart necessary and practical knowledge of components of Internet of Things and develop skills required to build real-life IoT based projects.													
Course Outcomes: Upon successful completion of this course, students should be able to:														
CO1	Understand internet of Things and its hardware and software components.													
CO2	Interface I/O devices, sensors & communication modules.													
CO3	Remotely monitor data and control devices.													
CO4	Develop real life IoT based projects.													
CO5														
CO6														
CO-PO & PSO Mapping														
COs	PROGRAMME OUTCOMES													
	1	2	3	4	5	6	7	8	9	10	11	12		
CO1	3	0			1							1		
CO2	2	3	1	3	2							1		
CO3	2	3	3	3	2				1		1	2		
CO4	2	3	3	3	2				1		1	2		
CO5	2	2	3											
CO6	1	1	3											
Avg.	2	2	2.6	3	1.75				1		1	1.5		
SYLLABUS														
UNIT:1 (8 Hours) Introduction to IoT: - Architectural Overview, Design principles and needed capabilities, IoT Applications, Sensing, Actuation, Basics of Networking, M2M and IoT Technology Fundamentals- Devices and gateways, Data management, Business processes in IoT, Everything as a Service(XaaS), Role of Cloud in IoT, Security aspects in IoT. Domain-Specific IOTs: Home Automation, Cities, Environment, Energy, Retail, Logistics, Agriculture, Industry, Health & Life Style.														
UNIT:2 (9 Hours) Elements of IoT: - Hardware Components- Computing (Arduino, Raspberry Pi), Communication, Sensing, Actuation, I/O interfaces. Software Components- Programming API's (using Python/Node.js/Arduino) for CommunicationProtocols-MQTT, ZigBee, Bluetooth, CoAP, UDP, TCP.														
UNIT:3 (18 Hours) IoT Application Development: - Solution framework for IoT applications- Implementation of Device integration, Data acquisition and integration, Device data storage- Unstructured data storage on cloud/local server, Authentication, authorization of devices.														
UNIT:4 (10 Hours) IoT Case Studies: - IoT case studies and mini projects based on Industrial automation, Transportation, Agriculture, Healthcare, Home Automation.														

UNIT-5	(10 Hours)
IoT Data Management and Analytics: Data generation, acquisition, and processing in IoT, IoT data storage and management: Cloud storage, edge computing, and big data, Data analytics for IoT: Machine learning, predictive analytics, and real-time processing, Integration with cloud platforms: AWS IoT, Azure IoT, Google Cloud Io, Visualization and dashboards for IoT data.	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books: 1. VijayMadiseti, Arshdeep Bahga, "Internet of ThingsA Hands-On-Approach", 2014, ISBN:978 0996025515. 2. Dr. SRN Reddy, Rachit Thukral and Manasi Mishra, "Introduction to Internet of Things: A practical Approach", ETI Labs. 3. Pethuru Raj and Anupama C. Raman, "The Internet of Things: Enabling Technologies, Platforms, and Use Cases", CRC Press. 4. Jeeva Jose, "Internet of Things", Khanna Publishing House, Delhi.	
Reference Books: 5. AdrianMcEwen, "Designing the Internet of Things", Wiley Publishers, 2013, ISBN:978-1-118-43062-0 . 6. Daniel Kellmereit, "The Silent Intelligence: The Internet of Things". 2013, ISBN:0989973700 7. Adrian McEwen, "Designing the Internet of Things", Wiley. Raj Kamal, "Internet of Things: Architecture and Design", McGraw Hill. 8. Cuno Pfister, "Getting Started with the Internet of Things", O Reilly Media.	
e-Books: 9. Designing for the Internet of Things, O'Reilly, February 2015. 10. Getting Started with the Internet of Things, O'Reilly 4th edition, October, 2015.	

VII Semester [Fourth Year]
B. Tech in Computer Science and Engineering (2023 Admission)

SN	CC	Sub Code	Name of the Subject	L	T	P	C	ISA	ESA
THEORY									
1	PC	23BCSPC47001	Computer Graphics	3	0	0	3	40	60
2	PC	23BCSPC47002	Data Communication	3	0	0	3	40	60
3	PE	23BCSPE470**	Professional Elective-V (From Bucket)	3	0	0	3	40	60
4	HS	23BHSHS47001	Entrepreneurship Development	3	0	0	3	40	60
5	OE	23BCSPE470**	Open Elective-III (From Bucket)	3	0	0	3	40	60
			TOTAL	15	0	0	15	200	300
PRACTICAL/SESSIONAL									
5	PC	23BCSPC47102	Data Communication Lab	0	0	2	1	70	30
5	PST	23BCSPW47101	Major Project Work-I	0	0	6	3	50	50
6	PST	23BCSPW47102	Summer Industry Internship-III	0	0	0	1	0	50
			TOTAL	0	0	8	5	120	80
			TOTAL	15	0	8	20	320	380

BUCKETS FOR PROFESSIONAL ELECTIVES

Sl. No.	Semester	Professional Elective V
1	23BCSPE47011	Cryptography and Network Security
2	23BCSPE47012	Information Security
3	23BCSPE47013	Database Security

BUCKETS FOR OPEN ELECTIVES

Sl. No.	Semester	Open Elective III
1	23BCSPE47011	Data Analytics
2	23BCSPE47012	Computer Vision
3	23BCSPE47012	Intrusion Detection

Subject Code		Subject Code								L	T	P	C			
		Computer Graphics								3	0	0	3			
Course Educational Objectives																
CEO1	Discuss various algorithms for scan conversion and filling of basic objects and their comparative analysis.															
CEO2	Introduce the use of geometric transformations on graphics objects and their application in composite form															
CEO3	Impart frame extraction with different clipping algorithms and transformation to a graphics display device.															
CEO4	Introduce projections and visible surface detection techniques for display of 3D scene on 2D screen and rendering of projected objects to naturalize the scene in 2D view.															
Course Outcomes: : <i>Upon successful completion of this course, students should be able to:</i>																
CO1	To <i>explain</i> the structure of modern computer graphics systems and <i>implement</i> its primitives															
CO2	To <i>design, develop</i> and <i>model</i> key algorithms for modeling and .															
CO3	Apply Graphics in greater depth to more complex courses like Image Processing, Virtual, Augmented Reality, etc															
CO4	To <i>visualize</i> surface detection and Virtual reality for a better visual effect.															
CO5	Desing Illumination models and implement them.															
CO6	Implement Animations in real time applications															
CO-PO & PSO Mapping																
Cos	PROGRAMME OUTCOMES												PSOs			
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	
CO1	1	2														
CO2	2	1	1													
CO3	1	2	2													
CO4	1	3	1													
CO5	1	2														
CO6	2	1														
Avg.	1.3	1.8	2													
SYLLABUS																
UNIT:1 (10 Hours)																
Overview of Graphics System: Video Display Units, Raster-Scan and Random Scan Systems, Graphics Input and Output Devices, monitor and work station. Graphics output Primitives: point and line, drawing Algorithms: DDA and Bresenham’s, Circle drawing Algorithms: Midpoint Circle and Bresenham’s filled area primitives.																
UNIT:2 (12 Hours)																
Two Dimensional Geometric Transformations: Translation, rotation, Scaling, Reflection, Shear, Matrix Representation, Composite Transformations, Transformation between coordinate systems. Window-to-View port Coordinate Transformation. Line Clipping (Cohen-Sutherland Algorithm) and Polygon Clipping (Sutherland Hodgeman Algorithm).																
Three Dimensional Geometric and Modeling Transformations: Translation Rotation, Scaling, Reflections, shear, Composite Transformation. Projections: Parallel Projection and Perspective Projection.																

UNIT:3	(10 Hours)
Object representation: Spline Representation, Bezier Curves and B-Spline Curves. Fractal Geometry: Fractal Classification and Fractal Dimension. Visible Surface Detection Methods: Back-face Detection, Depth Buffer, A- Buffer, Scan- line Algorithm and Painters Algorithm.	
UNIT:4	(12 Hours)
Light pattern and Illumination Models: Aliasing and Anti aliasing, Half toning, Thresholding and Dithering, Basic Models, Displaying Light Intensities. Surface Rendering Methods: Polygon Rendering Methods: Gouraud Shading and Phong Shading.	
UNIT-5	[10 Hours]
Computer Animation: Types of Animation, Key frame Vs. Procedural Animation, methods of controlling Animation, Morphing. Virtual Reality: Types of Virtual reality systems, Input and Output Virtual Reality devices	
Teaching Methods: Chalk& Board/ PPT	
Text Books:	
1. <i>Computer Graphics with Virtual Reality System, Rajesh K.Maurya, WileyDreamtech.</i> 2. <i>Computer Graphics, D. Hearn and M.P. Baker (C Version), Pearson Education</i>	
Reference Book:	
1. <i>Computer Graphics Principle and Practice , J.D. Foley, A.Dam, S.K. Feiner, Addison, Wesley</i> 2. <i>Procedural Elements of Computer Graphics- David Rogers (TMH)</i> 3. <i>Computer Graphics: Algorithms and Implementations – D.P Mukherjee &Debasish Jana (PHI)</i> 4. <i>Introduction to Computer Graphics & Multimedia – AnirbanMukhopadhyay</i>	

Subject Code		Name of the Subject				L	T	P	C						
		Data Communication				3	0	0	3						
Course Educational Objectives															
Course Outcomes: <i>Upon successful completion of this course, students should be able to:</i>															
CO1	Understand the fundamental concepts and components of data communication and transmission media.														
CO2	Analyze and apply data link layer protocols and error handling techniques for reliable communication.														
CO3	Evaluate network layer protocols, routing algorithms, and IP addressing schemes.														
CO4	Implement transport layer protocols and manage end-to-end communication effectively.														
CO5	Identify and use application layer protocols for various data communication services.														
CO6	Explore advanced topics in data communication and assess their impact on modern network architectures.														
CEO1	To discuss the digital data communication techniques														
CEO2	Gain knowledge on basic concepts of data communication layers, protocols and performance														
CEO3	Understand a few representative protocols and network components														
CEO4	To introduce the functions of different layers from deployed examples														
COURSE EDUCATIONAL OBJECTIVES:															
CO-PO MAPPING															
Cos	PROGRAMMEOUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	3	1													
CO2	3	1													
CO3	2	2													
CO4	2	3		1											
CO5	2	1													
CO6	2	2													
Avg.	2.3	1.6		1											
SYLLABUS															
UNIT:1 (14 Hours)															
Introduction to Computer Networks, Components, Devices, Topology, OSI model, layers in OSI model, TCP/IP Model, Address types (Physical, Logical, Port, Application-Specific Addresses), Analog and Digital Signal. Transmission Media (Guided Media, Unguided Media).															
Physical Layer: line encoding, Block coding, Binary data to Digital Signal Conversion, Digital to Analog Signal Conversion, Analog to Analog Signal Conversion, Analogue to Digital Signal conversion. Digital Transmission types (Serial and Parallel), Transmission Modes (Simplex, Half-Duplex and Full-Duplex), Multiplexing (FDM, TDM, WDM, CDM).															
UNIT:2 (10 Hours)															
Switching and Types: Circuit Switched Networks, Datagram Networks, Virtual Circuit Networks.															
Data link layer: Introduction, Error Detection and Correction techniques (Checksum, Cyclic Redundancy Check, Hamming Code (1-bit error rectification)), Framing, HDLC Frame Structure and															

Types, HDLC (NRM, ABM, ARM), Point to Point Protocol, Flow and Error Control, Noiseless Channels protocols (Simplest, stop-and-Wait), Noisy Channels protocols (Stop-and-Wait ARQ, Go-Back-N ARQ, Selective Repeat ARQ),	
UNIT:3	(11 Hours)
Medium Access sub layer: Multiple Access - Random Access Protocols (ALOHA, CSMA (non-persistent, 1-persistent, p-persistent), CSMA/CD, CSMA/CA), Controlled Access Protocols (Reservation, Polling, Token Passing), Channelization techniques (FDMA, TDMA, CDMA).	
Network Layer: Internetworking, IP Address, IPv4(Header structure, Class types, subnetting), IPv6 (Header structure, Address structure), IPV4 to IPV6 Address Mapping, Address Resolution Protocol, ICMP.	
UNIT:4	(10 Hours)
Transport Layer: User Datagram Protocol, Transmission Control Protocol, Process-To-Process Delivery, Congestion Control, Quality of Service.	
Application Layer: Domain Name System - Address Resolution, TELNET, E-MAIL (SMTP, POP3), FTP, HTTP, HTTPS, Voice Over IP, Firewall.	
Unit-5	(10 Hours)
Application Layer: Domain Name System - Address Resolution, TELNET, E-MAIL (SMTP, POP3), FTP, HTTP, HTTPS, Voice Over IP, Firewall	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	

Text Book:
1. <i>Data Communications and Networking – Behrouz A. Forouzan, Fourth Edition TMH,2006.</i> 2. <i>Computer Networks -- Andrew S Tanenbaum, 4th Edition, Pearson Education.</i>
Reference Book:
1. <i>An Engineering Approach to Computer Networks-S.Keshav,2nd Edition, Pearson Education</i> 2. <i>Understanding communications and Networks,3rd Edition, W.A.Shay, Cengage Learning.</i> 3. <i>Computer and Communication Networks ,Nader F. Mir, Pearson Education</i> 4. <i>Computer Networking: A Top-Down Approach Featuring the Internet, James F.Kurose, K.W. Ross, 3rd Edition, Pearson Education.</i>

Subject Code	Subject Code										L	T	P	C	
	Cryptography and Network Security										3	0	0	3	
Course Educational Objectives															
CEO1	Provide an Overview of Security Architecture and Number Theory used in Cryptography.														
CEO2	Introduce and Explain Different Security Algorithms and their Implementation.														
CEO3	Demonstrate Real-Time Security Threats & Security Types.														
CEO4	Identification of different key Security Issues in Email, IP & Web Security and Its Solution.														
Course Outcomes: : Upon successful completion of this course, students should be able to:															
CO1	Explain and use the concept of Number Theory in Cryptography.														
CO2	Design and implement of various Cryptography Algorithms.														
CO3	Interpret and Compare Different types of Authentication Application.														
CO4	Identify and Examine Key Security Issues in Email, IP and Web security.														
CO5															
CO6															
CO-PO & PSO Mapping															
Cos	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	3	2													
CO2	3	2	2	3											
CO3	2	2	3	2											
CO4	3	2	2												
CO5	1	3	1												
CO6	2	1	1												
Avg.	2.3	2	1.8	2.5											
SYLLABUS															
UNIT:1 (10 Hours)															
INTRODUCTION & NUMBER THEORY: Services, Mechanisms and attacks-the OSI security architecture-Network security model-Classical Encryption techniques (Symmetric cipher model, substitution techniques, transposition techniques, steganography).FINITE FIELDS AND NUMBER THEORY: Groups, Rings, Fields-Modular arithmetic-Euclid's algorithm-Finite fields- Polynomial Arithmetic –Prime numbers-Fermat's and Euler's theorem-Testing for primality-The Chinese remainder theorem- Discrete logarithms.															
UNIT:2 (14 Hours)															
BLOCK CIPHERS & PUBLIC KEY CRYPTOGRAPHY: Data Encryption Standard-Block cipher principles-block cipher modes of operation-Advanced Encryption Standard (AES)-Triple DES-Blowfish-RC5 algorithm. Public key cryptography: Principles of public key cryptosystems-The RSA algorithm-Key management - Diffie Hellman Key exchange-Elliptic curve arithmetic-Elliptic curve cryptography.															
HASH FUNCTIONS AND DIGITAL SIGNATURES: Authentication requirement– Authentication function – MAC – Hash function – Security of hash function and MAC –MD5-SHA-HMAC–CMAC-Digital signature and authentication protocols–DSS–ElGamal–Schnorr.															
UNIT:3 (12 Hours)															
SECURITY PRACTICE & SYSTEM SECURITY: Authentication applications – Kerberos – X.509 Authentication services - Internet Firewalls for Trusted System: Roles of Firewalls – Firewall related terminology- Types of Firewalls - Firewall designs - SET for E-Commerce Transactions. Intruder															

Intrusion detection system – Virus and related threats – Countermeasures – Firewalls design principles – Trusted systems – Practical implementation of cryptography and security.

UNIT:4

(10 Hours)

E-MAIL, IP & WEB SECURITY: E-mail Security: Security Services for E-mail-attacks possible through E-mail - establishing keys privacy-authentication of the source-Message Integrity-Non-repudiation-Pretty Good Privacy-S/MIME. IP Security: Overview of IPSec - IP and IPv6- Authentication Header-Encapsulation Security Payload (ESP)-Internet Key Exchange (Phases of IKE, ISAKMP/IKE Encoding).

UNIT-5

(10 Hours)

Web Security: SSL/TLS Basic Protocol-computing the keys- client authentication-PKI as deployed by SSL Attacks fixed in v3- Exportability-Encoding-Secure Electronic Transaction (SET).

Teaching Methods: Chalk& Board/ PPT/Video Lectures

Text Books:

1. “*Cryptography and Network Security: Principles and Practice*” by William Stallings
2. *Cryptography and Network Security*”by Behrouz A.Forouzan,Debdeep Mukhopadhyaya

Subject Code		Name of the Subject								L	T	P	C	QP	
		Data Analytics								3	0	0	3	A	
Course Educational Objectives															
CEO1	Provide an insight into Data pre-processing, summarization, and visualization techniques.														
CEO2	Teach role of data analytics in business decision making.														
CEO3	Teach Model building and validation using various techniques.														
Course Outcomes: <i>Upon successful completion of this course, students should be able to:</i>															
CO1	Understand and build regression models and use them for prediction														
CO2	Analyze data to infer underlying patterns and formulate recommendations.														
CO3	Outline the scope and limitations of several state-of-the-art data analytics methods.														
CO4	Implement different data analytics models in a business environment.														
CO5															
CO6															
CO-PO & PSO Mapping															
COs	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	3	3													
CO2	3	3	2												
CO3	3	3	3												
CO4	3	3	3	2	2										
CO5															
CO6															
Avg.	3	3	2.6	2	2										
SYLLABUS															
UNIT:1 (10 Hours) Predictive Analytics: Linear Methods for Regression and Classification: Overview of supervised learning, Linear regression models and least squares, Multiple regression, Multiple outputs, Subset selection, Ridge regression, Lasso regression, Linear Discriminate Analysis, Logistic regression, Perception learning algorithm.															
UNIT:2 (10 Hours) Neural Networks (NN), Support Vector Machines (SVM) and K-nearest Neighbor: Fitting neural networks, Back propagation, Issues in training NN, SVM for classification, Reproducing Kernels, SVM for regression, K-nearest-Neighbor classifiers (Image Scene Classification).															
UNIT:3 (10 Hours) Unsupervised Learning and Random forests: Association rules, Cluster analysis, Principal Components, Random forests and analysis. Inferential Statistics and Prescriptive analytics Assessing Performance of a classification Algorithm (t-test, McNemar’s test, Paired t-test, paired F-test), Analysis of Variance, Creating data for analytics through designed experiments. Introduction to big data and Challenges for big data analytics.															
UNIT:4 (10 Hours) Implementation of following methods using R or Mat lab (One of the class tests with a weight age of 15 marks be used to examine these implementations): Simple and multiple linear regression, Logistic regression, Linear discriminate analysis, Ridge regression, Cross-validation and boot strap.															

fitting classification and regression trees, K-nearest neighbors, Principal component analysis, K-means clustering.

UNIT-5

(10 Hours)

Advanced Data Analytics and Applications Text Analytics and Natural Language Processing (NLP): Sentiment Analysis, Topic Modeling, Text Classification, Time Series Analysis: Forecasting Methods, ARIMA, Seasonal Decomposition, Anomaly Detection: Techniques and Applications, Ethical Considerations and Data Privacy: GDPR, Data Governance, Ethics in Data Analytics, Emerging Trends: Edge Analytics, Automated Machine Learning (AutoML), Data Analytics in IoT, AI-Driven Analytics

Text Books:

1. Trevor Hastie, Robert Tibshirani, Jerome Friedman , *The Elements of Statistical Learning- Data Mining, Inference, and Prediction*, Second Edition , Springer Verlag, 2009.
2. G. James, D. Witten, T. Hastie, R. Tibshirani- *An introduction to statistical learning with applications in R*, Springer, 2013.
3. E. Alpaydin, *Introduction to Machine Learning*, Prentice Hall Of India, 2010.

Reference Books:

1. C.M. Bishop – *Pattern Recognition and Machine Learning*, Springer, 2006
2. L. Wasserman- *All of statistics*

Subject Code				Subject						L	T	P	C			
				Entrepreneurship Development						3	0	0	3			
Course Educational Objectives																
CEO1		Explore the entrepreneurial mindset and culture that has been developing in companies														
CEO2		Examine the entrepreneurial process involved in both pursuing an entrepreneurial venture within a large company and the creating and managing a new enterprise for implementation of an entrepreneurial venture.														
CEO3		Discuss the dynamics of participating on a business team and the power inherent in a team relative to individual effort														
CEO4		Provide the background and tools necessary to understand and participate in the entrepreneurial process within a large company, in a new venture or as an investor														
Course Outcomes																
CO1		Understand the concepts of entrepreneurship and skills required to be an entrepreneur														
CO2		Identify the Business Opportunities to and evaluate the feasibilities of the Project;														
CO3		Understand the formalities for setting up of a new venture and know the concepts of														
CO4		Evaluate different financial institutions supporting small scale industry.														
CO5		Develop skills of managing Finance, Marketing and HR of a business organization														
CO6		Analyze the impact of emerging trends in entrepreneurship on the global business														
CO-PO & PSO Mapping																
COs		PROGRAMME OUTCOMES												PSOs		
		1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1		1	2				3									
CO2		2	1				0		3			3				
CO3		1	2				0			2		2				
CO4		1	1				0					3				
CO5		1	1				0			3						
CO6		2	1													
AVG		1.3	1.3				0.6		0.6	2.5		1.6				
SYLLABUS																
Unit – I [10Hrs]																
Concept of Entrepreneurship, Motivation for Economic Development and Entrepreneurial Achievement, Enterprise and Society Why and how to start Business – Entrepreneurial traits and skills, Mind Vrs Money in Commencing New Ventures, Entrepreneurial success and failures, Environmental dynamics and change. Entrepreneurial Process Step by step approach to entrepreneurial start up Decision for Entrepreneurial start up.																
Unit - II [10Hrs]																
Identifying the Business opportunity - Business opportunities in various sectors, formalities for setting up small enterprises in manufacturing and services, Environmental pollution and allied regulatory and non-regulatory clearances for new venture promotion in SME sector. Writing a Business plan, components of a B-Plan, determining Bankability of the project.																
Unit – III [10 Hrs]																
Central / State level Institution promoting SME. Financial Management in small business. Marketing Management, problems & strategies Problems of HRM – Relevant Labour – laws. Sickness in Small Enterprises. Causes and symptoms of sickness – cures of sickness. Govt. policies on revival of sickness and remedial measures.																

Unit – IV

Innovation and Growth Strategies: Innovation and Creativity in Entrepreneurship: Techniques (Design Thinking, Blue Ocean Strategy), Processes, and Best Practices, Strategic Planning for Growth: Market Penetration, Market Development, Product Development, Diversification, Scaling up a Business: Processes, Challenges, and Strategies, Mergers and Acquisitions: Concept, Types, Benefits, and Challenges
Case Studies: Successful Start-ups and Learnings from Failure

Unit – V

Social Entrepreneurship and Sustainable Development- Social Entrepreneurship: Definition, Importance, and Types, Role of Social Enterprises in Community Development, Sustainable Development Goals (SDGs) and Entrepreneurship, Green and Inclusive Entrepreneurship: Concepts, Practices, and Case Studies, Emerging Trends: Digital Entrepreneurship, Women Entrepreneurship, Rural Entrepreneurship

Teaching Methods: Chalk& Board/ PPT/Video Lectures/Lecture by Industry Expert/MOOCs

Text Book:

1. Robert Hisrich, & Michael Peters: Entrepreneurship, TMH, 2009. Dollinger: Entrepreneurship, Pearson, 2009.

Reference Book:

1. Agarwal: Indian Economy, WishwaPrakashan 2009.
2. Dutt&Sundaram: Indian Economy, S.Chand, 2009
3. B D Singh.: Industrial Relations & Labour Laws, Excel, 2009.
4. Aruna Kaulgud: Entrepreneurship Management by, Vikas publishing house, 2009.
5. Essential of entrepreneurship and small business management by Thomas W. Zimmerer & Norman M. Searborough, PHI-2009.
6. ND Kapoor: Industrial Law, Sultan Chand & Sons, 2009

VIII Semester [Fourth Year]
B. Tech in Computer Science and Engineering (2023 Admission)

SN	CC	Sub Code	Name of the Subject	L	T	P	C	ISA	ESA
THEORY									
1	PE	23BCSPE480**	Professional Elective-VI (From Bucket)	3	0	0	3	40	60
2	OE	23BCSOE480**	Open Elective-IV (From Bucket)	3	0	0	3	40	60
3	OE	23BCSOE480**	Open Elective-V (From Bucket)	3	0	0	3	40	60
			TOTAL	9	0	0	9	120	180

PRACTICAL/SESSIONAL									
4	PST	23BCSPW48101	Technical Seminar	0	0	2	1	0	50
5	PST	23BCSPW48102	Major Project Work-II	0	0	6	3	50	50
			TOTAL	0	0	8	4	50	100
			TOTAL	9	0	8	13	170	280

BUCKETS FOR PROFESSIONAL ELECTIVES

Sl. No.	Semester	Professional Elective VI
1	23BCSPE48011	Block Chain Technology
2	23BCSPE48012	Malware Analysis
3	23BCSPE48013	Cyber Law

BUCKETS FOR OPEN ELECTIVES

Sl. No.	Semester	Open Elective IV & V
1	23BCSOE48011	Real Time Systems
2	23BCSOE48012	Embedded Systems
3	23BCSOE48021	E-Commerce & ERP
4	23BCSOE8022	Cyber Security
5	23BCSOE87023	Digital Forensics

Subject Code		Name of the Subject								L	T	P	C		
		Malware Analysis								3	0	0	3		
Course Educational Objectives															
CEO1: Understand the fundamentals of Dynamic and Static analysis & Gain knowledge about running malware in a virtual environment.															
CEO2: Study about disassembly constructs and their structures.															
CEO3: Study about new processors and file types using the IDA SDK															
CEO4: Explore popular plug-ins that make writing IDA scripts easier, allow collaborative reverse engineering & understand how to best approach the subject of Android malware threats and analysis.															
Course Outcomes: <i>Upon successful completion of this course, students should be able to:</i>															
CO1	Gain knowledge about the different forms of malware.														
CO2	Set up a safe virtual environment to analyze malware & navigate, comment, and modify disassembly														
CO3	Use code graphing to quickly make sense of cross-references and function calls & Use IDA's built-in debugger to tackle hostile and obfuscated code.														
CO4	Learn procedures for recognizing and analyzing Android malware threats quickly and effectively.														
CO5	Implement Cross References using open sources														
CO6	Learn Behavioral and Code Analysis														
CO-PO & PSO Mapping															
Cos	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	3	2													
CO2	3	2	2												
CO3	3	3													
CO4	2	3													
CO5	1	1													
CO6	2	2													
Avg.	2.3	2.1	2												
SYLLABUS															
UNIT:1 (14 Hours)															
The Goals of Malware Analysis, Malware Analysis Techniques, Types of Malware, General Rules for Malware Analysis, Antivirus Scanning, Hashing- Fingerprint for Malware, Finding Strings, Packing Files, Detecting Packers with PEiD, Portable Executable File Format, Static, Runtime, and Dynamic Linking, Exploring Dynamically Linked Functions with Dependency Walker, Imported and Exported Functions, PotentialKeylogger.exe: An Unpacked Executable, Examining PE Files with PEview, Viewing the Resource Section with Resource Hacker, Using Other PE File Tools, PE Header Summary.															
UNIT:2 (12 Hours)															
The Structure of a Virtual Machine, Creating Your Malware Analysis Machine, Using Malware Analysis Machine, The Risks of Using VMware for Malware Analysis, Record/Replay: Running Your Computer in Reverse, Sandboxes: The quick-and-dirty Approach, Running Malware, Monitoring with															

Process Monitor, Viewing Processes with process Explorer, Comparing Registry Snapshots with Regshot, Faking a Network, Packet Sniffing with Wireshark, Using INetSim, Basic Dynamic tools in practice, Reverse-Engineering, The x86 Architecture, Recognizing C Code Construct in Assembl

UNIT:3

(12 Hours)

Disassembly Theory, The Why and how of Disassembly, Reversing and Disassembly tools, Getting started with IDA, IDA Data Displays, Disassembly Navigation, Disassembly manipulation, Recognizing Data Structure Use, Creating IDA Structures, Using Structure Templates, Importing New Structures, Using Standard Structures, IDA TIL Files, C++ Reversing Primer- The this Pointer, Virtual Functions and Vtables, The Object Life Cycle, Name Mangling, Runtime Type Identification, Inheritance Relationships, C++ Reverse Engineering References.

UNIT:4

(14 Hours)

Cross-References, Function Calls, IDA graphics, Console mode IDA, IDA's Batch Mode, Customizing IDA's, Library Recognitions, Augmenting Function Information, Augmenting predefined functions, The Infamous Patch Program Menu, IDA Output Files and Patch Generation, IDA Scripting, IDA Software Development Kit, The IDA Application Programming Interface, Writing a Plug-in, Plug-in User Interface Options, IDA Loader Modules, Processor Module Architecture, Real World Applications- Vulnerability Analysis, Introduction to the Android Operating System and Threat, Malware Threats, Hoaxes, and Taxonomy, Open Source tools.

UNIT-5

(10 Hours)

Behavioral and Code Analysis: Analyzing malware persistence mechanisms, Identifying and disabling malware communication channels, Understanding command-and-control (C2) infrastructures, Techniques for unpacking and decrypting malware.

Exploit Development and Analysis: Introduction to exploit development, Analyzing vulnerabilities and exploits, Understanding buffer overflows, heap sprays, and ROP chains, Case studies of famous exploits.

Teaching Methods: Chalk& Board/ PPT/Video Lectures

Text Books:

1. Michael Sikorski, *Practical Malware Analysis – The Hands-On Guide to Dissecting Malicious Software*, Kindle Edition, No Starch Press; 1 edition (1 February 2012), ISBN: 1593272901.
2. Chris Eagle, *The IDA Pro Book*, 2nd Edition, No Starch Press, 2011. ISBN-10: 1-59327- 2898.

Reference Books:

1. Ken Dunham, *Android Malware and Analysis*, Kindle Edition, Auerbach Publications. International Standard Book Number-13:978-1-4822-5220-0.

Subject Code	Name of the Subject										L	T	P	C	
	Cyber Security										3	0	0	3	
Course Educational Objectives															
CEO1: Learn to model the various types of threat															
CEO2: Understand the Information Assurance planning strategies.															
CEO3: Acquire knowledge by analyzing software systems.															
CEO4: Understand and apply different countermeasures and protect information & Perform vulnerability testing.															
Course Outcomes: : Upon successful completion of this course, students should be able to:															
CO1	Acquire the basic knowledge about the Information Assurance & Design an appropriate Policy for the organization.														
CO2	Deliver professional, ethical, legal, security and social issues and responsibilities in an effective manner.														
CO3	Develop risk management strategies for an enterprise.														
CO4	Provide the understanding of different security mechanisms used in various areas of computing & Apply the current technical concepts and practices in the core information technologies.														
CO5	Implement Security issues.														
CO6	Provide Information Assurance Awareness.														
CO-PO & PSO Mapping															
Cos	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	3	-	-	-	-	-	2	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	3	3	-	3	-	-	-	-	-	-	-	-	-	-	-
CO5	1	2													
CO6	2	2													
Avg.	2.5	2.5		3				2							
SYLLABUS															
UNIT:1 (14 Hours)															
Information Assurance Basics, The Need for Information Assurance, Key Information Security concepts, Critical characteristics of Information, MSR model, Security in System lifecycle, NIST Approach to Securing SDLC, Security Professionals and Organizations, Communities of Interest, Information Security: Is it an art or Science?, Information Assurance Concepts : Defense in Depth, Information Assurance in Cyber Security, CIA Triangle, The Need for Security, Categories of Threats, Software Attacks types, Other vulnerabilities, Implications from Lack of Information Assurance.															
UNIT:2 (12 Hours)															
Information Security Planning, Information Security Governance, Policy, Standards and Practices, Policy Management, Information Security Blueprint, Continuity Strategies, Crisis Management, Information Asset Life Cycle, Plan, Do, Check, Act Model, Current Practices : Due Care and Due Diligence, Specific Laws and Regulations, International Laws and Acts, Standards and Best Practices, Plans for Information Assurance Strategy, Cryptology, Cipher methods, Cryptographic algorithms, Cryptographic tools, Protocols for secure communications.															

UNIT:3	(12 Hours)
Information Assurance Process : Managing Information Assurance, Information Security project management, Technical aspects of implementing Information Security, Non-Technical aspects of implementing Information Security , Structure of an Information Assurance, Organizational Maturity, Asset Management , APM Maturity mode, Overview of Risk Management, Risk Identification, Risk Assessment, Risk control, Quantitative vs Qualitative Risk management practices, Recommended risk control practices, Process , Secure design through threat modeling, Importance of Policy, Information Assurance Policy, Policy Development Steps.	
UNIT:4	(10 Hours)
Benefits of Incorporating Security Considerations, System Development Life Cycle, Information Assurance in System Development Life Cycle, Information Assurance in the Service Acquisition Life Cycle, Physical and Environmental Security Controls, Handling of Media.	
UNIT-5	(10 Hours)
Information Assurance Awareness ,Training, and Education (AT and E), Purpose, Benefits, AT and E : Design, Development, AT and E : Assessment, Types of Learning Programs, Employment Policies and Practices, Security considerations for temporary employees, consultants and other workers , Preventive Information Assurance Tools, positioning and staffing the security function, Credentials for Information Security Professionals, Access control benefits, Access control Techniques, Administration.	
Teaching Methods: Chalk& Board/ PPT/Video Lectures	
Text Books: 1. Michael E. Whitman and Herbert J. Mattord, “Principles of Information Security”, 5th edition, 2015, Thomson Publications, ISBN 1111899134. 2. Steven Hernandez, Corey Schou, “Information Assurance Handbook: Effective Computer Security and Risk Management Strategies”, 1st Edition, 2014, McGraw Hill Osborne Media, ISBN: 0071821651, ISBN : 9780071821650.	
Reference Books: 1. William Stallings, “Cryptography and Network Security- Principles and Practice”, 6th Edition, 2013, Pearson, ISBN: 9780136073734.. 2. Corey Schou, Dan Shoemaker, “Information Assurance for the Enterprise”, Tata McGraw -Hill Edition, 2.	

Subject Code	Name of the Subject										L	T	P	C	
	Real-Time Systems										3	0	0	3	
Course Educational Objectives															
CEO1: Learn to model the various types of Real Time Systems															
CEO2: Understand the Hardware Requirements.															
CEO3: Acquire Languages on Real Time Systems															
CEO4: Understand and apply different RTS.															
Course Outcomes: : <i>Upon successful completion of this course, students should be able to:</i>															
CO1	Explain the fundamentals of Real time systems and its classifications.														
CO2	Understand the concepts of computer control and the suitable computer hardware requirements for real-time applications.														
CO3	Describe the operating system concepts and techniques required for real time systems.														
CO4	Develop the software algorithms using suitable languages to meet Real time applications.														
CO5	Apply suitable methodologies to design and develop Real-Time Systems.														
CO6	Design RTS and Development Methodologies.														
CO-PO & PSO Mapping															
Cos	PROGRAMME OUTCOMES												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	3	-	-	-	-	-	2	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	3	3	-	3	-	-	-	-	-	-	-	-	-	-	-
CO5	1	2													
CO6	2	2													
Avg.	2.5	2.5		1				1							
SYLLABUS															
UNIT:1 (14 Hours)															
Introduction to Real-Time Systems:															
Historical background, Elements of a Computer Control System, RTS- Definition, Classification of Real-time Systems, Time Constraints, Classification of Programs.															
Concepts of Computer Control:															
Introduction, Sequence Control, Loop Control, Supervisory Control, Centralized Computer Control, Hierarchical Systems.															
UNIT:2 (12 Hours)															
Computer Hardware Requirements for Real-Time Applications: Computer Hardware Requirements for Real-Time Applications: Introduction, General Purpose Computer, Single Chip Microcomputers and Microcontrollers, Specialized Processors, Process-Related Interfaces, Data Transfer Techniques, Communications, Standard Interface															

UNIT:3**(12 Hours)****Languages for Real-Time Applications:**

Introduction, Syntax Layout and Readability, Declaration and Initialization of Variables and Constants, Cutlass, Modularity and Variables, Compilation of Modular Programs, Data types, Control Structures, Exception Handling, Low-level facilities, Co-routines, Interrupts and Device Handling, Concurrency, Real-Time Support, Overview of Real-Time Language.

UNIT:4**(10 Hours)****Operating Systems:**

Introduction, Real-Time Multi-Tasking OS, Scheduling Strategies, Priority Structures, Task Management, Scheduler and Real-Time Clock Interrupt Handler, Memory Management, Code Sharing, Resource Control, Task Co-Operation and Communication, Mutual Exclusion.

UNIT-5**(10 Hours)****Design of RTS- General Introduction:**

Introduction, Specification Document, Preliminary Design. Single-Program Approach, Foreground/Background System.

RTS Development Methodologies:

Introduction, Yow-don Methodology, Ward and Mellor Method, Hatley and Pirbhai Method

Teaching Methods: Chalk& Board/ PPT/Video Lectures

Text Books:

- *Real-Time Computer Control, Stuart Bennet, 2nd Edn. Pearson Education. 2008.*

Reference Books:

1. "Real-Time Systems", C.M. Krishna, Kang G Shin, McGraw-Hill International Editions, 1997.
2. Real-Time Systems Design and Analysis, Phillip. A. Laplante, second edition, PHI, 2005.
3. Embedded Systems, Raj Kamal, Tata McGraw Hill, India, third edition, 2005.



A handwritten signature in blue ink, appearing to read "K. Murali Gopal".

Dr. K.MURALI GOPAL
HOD, CSE
GIET UNIVERSITY
GUNUPUR-765022